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Scientific background document

AI supported MCM development

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Executive summary

Artificial Intelligence (AI) is significantly advancing **medical countermeasures** (MCM) by speeding up research and development processes, particularly in medicine development. AI is increasingly used throughout the MCM lifecycle, including drug discovery, clinical development, and improving clinical trial efficiency, quality control, and regulatory compliance. The EU4Health programme 2021-2027 is paving the way for a strong European Health Union, including among its key priorities reinforcing the European Union's (EU) digital transformation and investing and promoting research and technologies through the 'Pharmaceutical Strategy for Europe'.

Following the COVID-19 pandemic, the European Commission (EC) created the **European Health Emergency Preparedness and Response Authority** (HERA) to enhance the EU's ability to manage health emergencies. Established in 2021, HERA collaborates with Member States, industry, and civil society to **improve preparedness and response to health threats**, focusing on the development, manufacturing, and equitable distribution of MCM. Key priorities include rapid characterisation of emerging pathogens, expedited development of vaccines, and leveraging AI to enhance drug development and clinical trials. HERA also aims to proactively identify innovations in MCM, support Member States in vaccine and therapeutic development, and increase funding for high-risk R&D projects through the HERA Invest initiative, utilising its expertise in biotechnology and biomanufacturing.

In recent years, AI has emerged as a powerful tool for **streamlining drug development and enhancing vaccination efficacy**. Recognised since before the COVID-19 outbreak, AI's potential in preparing for and responding to health emergencies was already being applied in some areas of MCM development. However, the pandemic marked a pivotal moment for the design and deployment of AI-based tools. AI technologies are now being used to analyse small compounds, peptides, and epitopes to create new therapeutics and guide modelling and screening processes. Machine learning (ML) techniques can leverage existing data to support drug repurposing by screening libraries of existing drugs or chemical compounds that can match with the target identified as well as contribute to vaccine advancement. Additionally, AI models play a crucial role in identifying and evaluating promising therapeutic targets and antigenic peptides that trigger robust immune responses, and in facilitating the design of novel drugs. This includes the development of small-molecule drugs and *de novo* protein therapeutics, where AI accelerates the drug discovery process by predicting molecular interactions and generating innovative therapeutic candidates from scratch. AI-driven solutions have also been shown to support tackling challenges associated with clinical trial design, optimisation, and ensuring the safety and efficacy of treatments. These models can enhance trial outcomes by optimising patient recruitment, improving selection criteria, and streamlining trial protocols.

This report examines how AI can enhance the development of MCM by accelerating research and development processes, thereby **facilitating drug discovery and uncovering innovative solutions** that traditional methods might overlook. Furthermore, the report discusses how AI can **improve the efficiency of clinical trials,** particularly to reduce the time needed to make new MCMs available during health emergencies. This includes streamlining patient recruitment, optimising trial design, analysing historical trial data for new mechanisms of action, and using real-time analytics to monitor treatment efficacy and safety. Additionally, the report includes evidence on how **AI can support decision-making on MCM development during health emergencies.** AI-driven algorithms can analyse critical data, aiding governments and MCM developers in effectively allocating funding and expediting the development of essential medical countermeasures. This approach ensures rigorously tested MCM and evidence-based decision-making. Finally, this report aims to assist the HERA in identifying areas for improvement in the strategic coordination of MCM development and enhancing the industrial capacity to produce and supply these countermeasures at the EU level.

AI-based tools have been identified through a **structured literature review and consultation activities, including in-depth interviews** addressed to relevant stakeholders from academia, industry and governmental sectors. The results are structured in six functional sections: 1) AI for drug discovery and drug repurposing; 2) AI for drug target identification; 3) AI for pre-clinical research; 4) AI for clinical research; 5) AI for vaccine development, and 6) AI to support decision-making on MCM development.
The literature review on AI-based tools highlights significant advancements in **drug discovery and repurposing**. In drug discovery, AI technologies such as ML, generative models and graph neural networks have been shown to **enhance the efficacy and accuracy of identifying crucial disease-related proteins**, leading to more targeted therapeutic strategies. AI's capacity to analyse large datasets also facilitates the discovery of new pharmaceutical targets by **pinpointing specific biomarkers and mutations**, which is especially valuable during health crises. For drug repurposing, advanced computational algorithms predict 3D structures of drug targets and identify drug interaction, **expediting the discovery of new uses** for existing drugs and **mitigating development risks** due to pre-established safety profiles. Techniques such as network-based approaches and structure-based methodologies further enhance these efforts by integrating diverse data sources and leveraging 3D structural information to design *de novo* drugs and small molecules.

The literature review of **AI-based tools for target identification** revealed that it is a notably complex process which involves various scientific fields, such as computational chemistry and chemoinformatics. ML strategies have been crucial in this domain, enabling the identification of unique and *de novo* compounds within extensive databases. Evidence has shown that deep learning technology is highly used for accurately **predicting a drug's affinity for specific targets**, thereby expediting the medication formulation process and **providing superior performance** compared to traditional methods. Additionally, AI and ML techniques have notably contributed to drug-target identification (DTI), with significant progress in response to emerging infectious diseases such as COVID-19. For instance, deep learning models were employed to screen large libraries of molecules to determine their efficacy against SARS-CoV-2, highlighting the potential of these technologies to **accelerate the discovery of new compounds**. Furthermore, structure-based approaches, such as molecular docking and the Retro Drug Design workflow, have shown effectiveness in predicting **drug-target interactions**. These integrated methods use computational power to prioritise promising candidates, predict binding affinities and nodes, and explore molecular structures with high efficiency and atomic accuracy, thereby significantly accelerating drug discovery.

AI has also contributed to transforming **pre-clinical drug development** by significantly streamlining the target identification and validation processes. Evidence suggests that AI algorithms can process extensive genetic and protein interaction data, allowing them to prioritise promising targets and **predict potential side effects**. This capability not only accelerates the development timeline but also conserves valuable resources. Moreover, AI plays a critical role in evaluating the potential efficacy and toxicity of drug candidates before human trials. By analysing historical data from clinical trials and preclinical research, AI can **identify patterns that predict a drug's effectiveness** and possible adverse effects, thereby enabling researchers to focus on candidates with the highest likelihood of success and minimal risk. In addition, AI techniques have been deemed instrumental in developing MCM, such as surveillance and diagnostic tools. By leveraging AI for rapid pathogen detection and analysis, researchers can enhance times and diagnostic accuracy, ultimately improving public health preparedness and response strategies.

The literature review of AI-based tools for **clinical research** highlighted that AI can significantly **boost both the efficiency and effectiveness of clinical** trials by improving patient matching, optimising treatment protocols, and predicting patient outcomes. However, AI also has the potential to influence bias in clinical trials, both positively and negatively. Addressing the digital divide and **implementing robust privacy safeguards are essential for improving the diversity of trial cohorts and ensuring equitable participation**. Furthermore, selecting the right patient population is a critical component of

medical research and treatment. AI experts mention that the latter can be addressed by refining precision in patient matching, increasing trial diversity, and enhancing overall trial efficiency.
Finally, evidence on how AI can support vaccine development during health emergencies is mainly focused on COVID-19 vaccines where, according to the experts, the availability of pre-existing vaccines for SARS and MERS played a crucial role in the rapid adaptation to SARS-CoV-2, underscoring the importance of adaptable technologies for responding to future pandemics. Recent advancements in ML, including ***in silico*** **techniques and neural networks, have shown considerable promise in predicting the structures of antigens**, which could significantly aid in the development of new vaccines. In addition, ML algorithms have shown to be particularly effective at **identifying specific immune response targets**, predicting vaccine efficacy, and modelling antigen-antibody interactions not only for well-known pathogens but also for those emerging pathogens which do not have associated structures, and ML can support vaccine development. These capabilities enhance the efficiency and precision of vaccine development, leading to **more targeted and effective** solutions.

Overall, this study finds that adopting AI-based tools to support MCM development can enhance the EU's crisis preparedness and response capabilities by optimising drug development and improving the efficiency of clinical trials and pre-clinical research. In addition, AI significantly reduces time and cost by streamlining intensive processes, cutting down development timelines and lowering R&D expenditures. Additionally, AI enhances safety and efficacy through advanced analysis and prediction of adverse effects, creating safer drugs and vaccines, thereby improving the chances of regulatory approval. These improvements are crucial for advancing precision medicine and enabling more personalised and effective treatment strategies tailored to the unique characteristics of individual patients and diverse population groups.

The study concludes that AI-based tools for MCM development can greatly improve the EU's crisis preparedness by automating processes to reduce costs and timelines, and streamlining clinical trials, by predicting failures early, minimising trial dropout rates, and improving safety and efficacy through advanced analysis, resulting in safer drugs and vaccines with higher regulatory approval potential.

Overall, AI accelerates innovation, improves therapeutic outcomes, and boosts resource efficiency in the pharmaceutical sector. The evidence reviewed underscores the importance of ensuring that AI applications throughout the medicinal product lifecycle comply with existing legal requirements, ethical principles, and fundamental rights. In the context of the HERA Work Plan 2024 the results of this work will inform the actions of the HERA to use AI to develop MCM before and during a health crisis.

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Acronyms and abbreviations

AI	Artificial Intelligence
ANN	Artificial Neural Network
CEPI	Coalition for Epidemic Preparedness Innovations
CGNN	Causal Generative Neural Network
CNN	Convolutional Neural Network
DMPNN	Directed Message-passing Neural Network
DT	Decision Tree
DTI	Drug Target Identification
DNA	Deoxyribonucleic Acid
DNN	Deep Neural Network
EC	European Commission
EMA	European Medicines Agency
EMRN	European Medicines Regulatory Network
EU	European Union
FDA	Food and Drug Administration
GAT	Graph Attention Network
GNN	Graph Neural Network
HERA	Health Emergency Preparedness and Response Authority
HIV	Human Immunodeficiency Virus
HMA	Heads of Medicines Agencies
HTS	High-throughput screening
K-NN	K-Nearest Neighbours
LLM	Large language models
MCM	Medical Countermeasure
MDR	Medical device regulation
MERS	Middle East Respiratory Syndrome
ML	Machine Learning
MS	Member States
NLP	Natural Language Processing
OECD	Organisation for economic Cooperation and Development
R&D	Research and Development
RDD	Retro Drug Design
RF	Random Forest
RNA	Ribonucleic Acid
SARS	Severe Acute Respiratory Syndrome
SME	Small-medium enterprise
SVM	Support Vector Machine
VS	Virtual Screening
WHO	World Health Organization

Glossary

Term	Description
Artificial Intelligence system	Machine-based system that, for explicit or implicit objectives, infers, from the input it receives, how to generate outputs such as predictions, content, recommendations, or decisions that can influence physical or virtual environments.
Compound screening	The process in drug discovery where large libraries of chemical compounds are tested to identify potential candidates that have the desired biological or pharmacological activity.
Deep learning	Subset of ML that uses multi-layered neural networks, called deep neural networks, to simulate the complex decision-making power of the human brain.
Drug Discovery	The use of artificial intelligence (AI) techniques and algorithms to accelerate and enhance the process of discovering new pharmaceutical drugs.
Drug Repurposing	To find new uses for existing drugs by analysing large datasets, speeding up the process of discovering new treatments through AI and algorithms.
Drug Target Identification	The process of identifying the specific biological molecules, such as proteins or genes, that a drug will interact with to produce a therapeutic effect.
Generative AI	Generative AI is a type of AI that generates images, text, videos, and other media in response to inputted prompts.
Good Manufacturing Practice	Good Manufacturing Practice (GMP) is a system for ensuring that products are consistently produced and controlled according to quality standards. It is designed to minimise the risks involved in any pharmaceutical production that cannot be eliminated through testing the final product. GMP covers all aspects of production from the starting materials, premises, and equipment to the training and personal hygiene of staff.
In silico	It is a branch of science, based on *in silico* experimentation, that aims at grouping, interpreting, and elucidating biologic events through the organization and understanding of data and information based on the use of tools from biology, computing, mathematics, and related areas.
In-vitro	It refers to experiments conducted in a controlled laboratory environment using cells, tissues, or biochemical processes outside of a living organism.
In-vivo	Process performed after in vitro experiments to assess how a drug behaves in a complex biological system.
Large Language Models	Large language models are unsupervised deep learning models of considerable volume, pre-trained on large amounts of data. The underlying transformer is a set of neural networks each consisting of an encoder and a decoder with self-attention capabilities. Encoders and decoders extract meanings from a sequence of text and understand the relationships between words and phrases in it and can process entire sequences in parallel.
Machine learning	Branch of AI and computer science that focuses on using data and algorithms to enable AI to imitate the way that humans learn, gradually improving its accuracy.
Medical Countermeasure	MCM include drugs, vaccines, biologics, sensors, diagnostics, personal protective equipment, wearables, and other technologies

	that can assist in the detection, diagnosis, prevention, or treatment of a chemical, biological, radiological, or nuclear threat.
Molecular Docking	Computational technique used in drug discovery to predict how a small molecule, such as a drug candidate, interacts with a target protein at the molecular level.
Natural Language Processing	Natural language processing (NLP) is a branch of AI that enables computers to comprehend, generate, and manipulate human language. NLP can interrogate the data with natural language text or voice.
Process optimisation	Process optimisation is the systematic approach of analysing and improving business processes to achieve maximum efficiency, effectiveness, and quality.
Post market analysis	The process of monitoring and evaluating a product's performance after it has been released to the market.

1. Introduction

1.1. Study scope

Artificial Intelligence (AI) is playing an important role in advancing medical countermeasures (MCM) by expediting research and development processes (R&D), especially when it comes to medicines development. A fast-growing use of AI has been applied in the lifecycle of discovery, clinical development and delivery of MCM. In addition, AI has contributed to enhancing the industry's efficiency, quality control and regulatory compliance. In this respect, the EU4Health programme 2021-2027 is paving the way for a strong European Health Union, including among its key priorities reinforcing the European Union's (EU) digital transformation and investing and promoting research and technologies through the 'Pharmaceutical Strategy for Europe' (DG SANTE, 2021a).

Following the onset of the COVID-19 pandemic, the European Commission (EC) established the European Health Emergency Preparedness and Response Authority (HERA) to address health emergencies by strengthening the EU's ability to prevent, detect and respond to serious cross-border health threats by ensuring the development, manufacturing, procurement, and equitable distribution of key MCM. Since its establishment in 2021, HERA has engaged with Member States (MS) and various stakeholders, including industry and civil society, to develop activities aimed at improving preparedness and response to health threats around MCM. One of HERA's key priorities is to enhance the EU's ability to rapidly characterize emerging pathogens and develop effective MCM, including expediting the development, production, and distribution of vaccines. In this context, the processes of drug development and clinical trials are critical. Leveraging AI in these areas can significantly increase efficiency, improve data analysis and understanding, and support decision-making during emergencies and preparedness efforts.

HERA aims to enhance its horizon scanning and foresight capacities to proactively identify emerging innovations and technologies within the field of MCM, which will enable a rapid adaptation and response to new health threats. Additionally, HERA seeks to support MS in developing vaccines, therapeutics, and other MCM, as well as to increase financing for high-risk R&D projects through the *HERA Invest initiative*. This initiative will be bolstered with additional funding to mobilise essential resources for innovative MCM development and production. HERA, positioned uniquely between threat intelligence and the development and manufacturing of MCM, is **acquiring extensive knowledge in the fields of biotechnology and biomanufacturing**.

For instance, through the EU4Health programme, HERA intends to establish the **European Vaccine Development Hub** to expedite the development, production, and distribution of vaccines by fostering collaboration and innovation. This decentralised hub will involve various centres of excellence, pooling critical knowledge and connecting it with available capacities for clinical trials. These capacities will be created within the partnership on pandemic preparedness, in collaboration with EU clinical trial networks. Another example is HERA's ongoing support for the **development and access to a wide range of MCM**, with a particular focus on those included in the list of critical MCM established in 2023 by MS and HERA. This support aims to enhance preparedness and response to serious cross-border health threats by funding late-stage clinical trials, clinical investigations, and performance assessments or similar studies.

In its 2024 work plan, HERA has committed to supporting innovation related to AI and medicines development, outlining actions to adopt AI-supported technologies in this specific field (HERA, 2023). To this end, HERA has commissioned a report on AI-supported MCM development to identify the role of AI-based tools in advancing medicine development and accelerating drug discovery relevant to health

emergencies as well as reporting how AI can enhance the efficiency of clinical trials in preparedness and response to health emergencies.

This report presents the results of a literature review and consultation activities focused on pinpointing the various areas within the development of MCM where AI has been either implemented or holds the potential to accelerate and transform medicine development by identifying new molecular structures of pharmacological interactions and uncovering innovative solutions that might elude traditional approaches and pinpointing promising drug candidates. Furthermore, the report also outlines how AI can contribute to the efficiency of clinical trials in emergency preparedness and response. For instance, by streamlining patient recruitment and optimising trial design, AI can accelerate the development and evaluation of MCM and improve and refine the clinical trial design to improve efficacy and safety using real-time analytics. Ultimately, this report provides an overview of how AI can streamline decision-making on MCM development and how AI-driven algorithms can support the analysis of critical data, aiding governments and MCM developers in allocating funding to critical research and expediting the targeted development of necessary MCM.

1.2. Study objectives

This document aims to provide a scientific background to support HERA's implementation of the 2024 Work Plan and its preparedness activities. Specifically, it will assist HERA in analysing **the role of AI** in advancing the development of medical countermeasures (MCM) and **accelerating medicines development**. It will also outline how AI can enhance the efficiency of clinical trials, thereby strengthening health preparedness and response resilience.

More specifically, this document has three main objectives:

1. To explore the role of AI in accelerating the development of medicines and MCM:
    - Map which areas of drug discovery and drug repurposing can be enhanced by AI solutions.
    - Identify how AI can enhance vaccine development, particularly in response to emerging health threats.

2. To identify how AI can support and optimise clinical trial design, particularly during health emergency preparedness and response:
    - Map which AI tools can assist and support clinical trial design and optimisation.
    - Identify areas of clinical trial design where AI tools can enhance efficiency and optimisation, as well as support patient selection and recruitment.
    - Identify AI tools and solutions that improve both pre-clinical and clinical research phases.

3. To investigate how AI-driven algorithms can support the analysis of critical data aiding governments and MCM developers in allocating funding to critical research, expediting the targeted development of key MCM.

1.3. Structure of the Report

The following section of the report provides an overview of the state-of-the-art on the use of AI tools in MCM development, particularly in health emergency and response. This section provides the reader with an understanding of the potential uses of AI-based tools in the field and the areas of potential implementation to accelerate MCM development. Additionally, it also covers the evidence describing how AI can support decision-making during MCM development through the analysis of historical data.
Section 2 presents thoroughly the results of the desk research which includes the findings from both the literature review and consultation activities. Findings are organized into eight main sections:

- AI for drug discovery and drug repurposing
- AI for drug target identification

- AI for pre-clinical research
- AI for clinical research
- AI for vaccine development
- AI for automation and optimisation
- Limitations and challenges for AI in MCM development
- AI to support decision-making on MCM development

Section 3 “Discussion and conclusions” is dedicated to the synthesis of the key findings and presents the major implications of the main results. This section also discusses the limitations faced when introducing AI technology to MCM development and suggests directions for future research.
Annex 1 includes the methodology followed in this report, detailing the key research questions, the literature search strategy, the data extraction methodology and the organisation of consultation activities.

1.4. Overview of the state-of-the-art

According to the World Health Organization (WHO), the field of health emergency preparedness and response refers to the knowledge and **capacities to effectively anticipate, respond and recover** from the impacts of likely, imminent or current hazardous events or conditions (PAHO, n.d.). In the European context, the WHO Regional Office for Europe has initiated the development of a new regional strategy and action plan to strengthen health emergency preparedness, response, and resilience in the WHO European Region, namely “Preparedness 2.0 (World Health Organization, 2024)”. Among its objectives, the strategy aims to promote access to MCM by ensuring that preparedness efforts focus on expedited research and development. This includes pre-negotiated benefit-sharing agreements, appropriate financing and regulatory instruments, scalable manufacturing platforms, and technology transfer agreements. Additionally, it seeks to establish emergency supply chains to enable **equitable access and support evidence-based decision-making** for preparing and responding to health emergencies.
AI plays an important role in advancing MCM by expediting research and development processes, especially when it comes to medicines development. AI is currently being used in a variety of uses for this purpose (Wong et al. 2023). For example, evidence shows that machine learning (ML) can discern patterns within molecular structures, pharmacological interactions, and biological responses, pinpointing promising drug candidates. It can also accelerate drug discovery, uncovering innovative solutions that might elude traditional approaches and can contribute to improving the efficiency of clinical trials to reduce the time needed to make MCM available during health crises.

With the spread of COVID-19, the need for faster and optimised drug development processes emerged as a major concern, both to meet the immediate demand for medicines able to treat COVID-19 and for the necessity of a vaccine that could contain the transmission and the symptoms in the shortest possible time. The spread of the pandemic contributed to an acceleration of AI-based techniques in this field, requiring the commitment of policymakers and developers to allocate funding to critical research for future needs (Qureshi et al. 2023).

Traditionally, drug discovery is a lengthy and challenging process, with few candidates successfully reaching clinical trials, posing significant economic risks for companies and research centres (Kim et al. 2023). However, in recent years, drug discovery has been radically transformed by novel analytical methods and computational advances powered by AI. AI has demonstrated its potential to revolutionize the development and discovery of new medicines, offering benefits in both accuracy and speed.

In the need to streamline the processes that lead to the development of new medicines, AI-based methods have been widely used, proving to be able to develop new bioactive compounds and molecules thanks to their ability to process and cross-reference a vast amount of data through very varied techniques, such as deep learning (Syrowatka et al., 2021). It is estimated that in the coming years, AI will enhance its intervention capabilities, despite some limitations. Therefore, it will be necessary to

invest in AI to fully exploit its potential, preparing for future pandemics or health emergencies where there may be an urgent need for new medicines. Combining the predictive power of AI with the expertise of human researchers, will enhance the optimisation of the drug discovery process and accelerate the development of new medicines (Hutson, 2024).

AI tools are specifically used to accelerate various stages of the drug discovery pipeline, a highly complex and intricate process which includes the following phases (Choudhury et al. 2022):

- **Target Identification**: AI can help to identify novel targets by analysing a vast array of proteins and their associations with diseases.
- **Compound Screening**: Generative AI can be used to design *de novo* molecules, proteins or small molecules, tailored for precise specifications or targets.
- **Clinical Development**: AI can predict drug responses to enhance clinical development.
- **Molecule Optimisation**: Reinforcement learning optimises molecules to achieve desired properties.
- **Pre-clinical Development**: AI can assist in patient sampling, biomarker identification, and endpoint prediction.
- **Post-market Analysis**: AI can detect adverse effects, interactions, and potential for drug repurposing during the review and post-market safety phases.

In all activities included within the pipeline of developing a new drug, the capabilities of AI to process large amounts of data come into play. For example, for target identification ML algorithms can analyse various types of data, including gene expression profiles, protein interaction networks, and genomic and proteomic data, to identify potential targets that are likely to be involved in disease pathways; for the optimisation of compounds, AI models can be trained using various types of data, such as known protein-ligand complexes, structural information, and molecular descriptors. Finally, in the post-market analysis, Natural Language Processing (NLP) can be used to mine scientific literature articles to report adverse effects. Overall, evidence has shown the availability of vast amounts of chemical and biological databases on which AI tools are trained which can support the development of active learning approaches where synthetic or experimental large datasets can be produced to refine existing generative models (Qureshi et al. 2023).

Examples of AI tools include a ML model developed by the University of Melbourne, designed to accurately predict antigenic changes in circulating influenza viruses and their ability to evade immunity from previous infections or vaccinations. This model analyses the genetic sequence of viral strains and incorporates data from past influenza seasons to forecast the results of antigenic lab tests. These tests are essential for assessing how a viral strain will respond to the current influenza vaccine, determining whether a vaccine update is needed, and identifying the most suitable vaccine strain if necessary (Shah et al. 2024). Another example is the Machine-learning Assistant Influenza Vaccine Strain Selection framework, which aims to drastically reduce the time required to select flu strains for annual vaccines, from months to just a few days (Gao et al. 2024). Another context of interest when it comes to health emergency is antimicrobial resistance. AI has been utilized to create new antibiotics and produce synergistic combinations of drugs. Machine learning algorithms examine data patterns related to antimicrobial use and resistance, allowing predictions about which microorganisms might develop resistance to specific drugs. This information can assist healthcare providers and policymakers in making informed choices regarding drug selection and usage (Ali, Ahmed, Aslam 2023).

An important activity on which AI is playing a prominent role is in clinical trial design and optimisation. Improving clinical research is a significant area of interest in the pharmaceutical industry, and AI can help organizations and pharmaceutical companies automate specific tasks. For instance, in trial design, AI can predict the likelihood of a trial's success based on the drug molecule, target disease, and patient eligibility criteria. During patient recruitment, which can take up to one-third of the study duration, AI can assist by refining eligibility criteria, predicting potential dropouts, and monitoring ongoing interventions.

The AI input not only streamlines the time-consuming phases but also contributes to the robustness of the trials (Sreepadmanabh et al. 2020).

The European Medicines Agency (EMA) and the Heads of Medicines Agencies (HMA) have published an AI work plan from 2024 to 2028, setting out a collaborative and coordinated strategy between stakeholders to maximise the benefits of AI while managing the risks of AI cutting-edge technology (EMA 2023). The work plan will help support the European Medicines Regulatory Network (EMRN) to embrace the opportunities of AI for productivity, automating processes and systems, increasing insights into data and supporting more robust decision-making to benefit public and animal health. The work plan is aimed at addressing four critical AI dimensions:

- guidance, policy and product support.
- efficiency and data protection of AI tools and technology
- collaboration and training
- research to accelerate learning and gain new insights.

As part of pharmaceutical data intelligence, AI is rapidly and profoundly moving into each field of medicine discovery and development. In addition, in 2023, EMA published a reflection paper outlining the current thinking on the use of AI to support safe and effective development, regulation and use of human and veterinary medicines. The paper reflects on principles relevant to the application of AI and ML at any step of a medicine's lifecycle. It highlights that a human-centric approach should guide all development and deployment of AI and ML. The use of AI in the medicinal product lifecycle should occur in compliance with the existing legal requirements, considering ethics and ensuring due respect to fundamental rights.

This report examines how AI can enhance the development of MCM by accelerating research and development processes, thereby **facilitating drug discovery and uncovering innovative solutions** that traditional methods might overlook. Furthermore, the report discusses how AI can **improve the efficiency of clinical trials,** particularly during health emergencies**.** This includes streamlining patient recruitment, optimising trial design, analysing historical trial data for new mechanisms of action, and utilizing real-time analytics to monitor treatment efficacy and safety. Additionally, the report includes evidence on how **AI can streamline decision-making on MCM development during health emergencies.** AI-driven algorithms can analyse critical data, aiding governments and MCM developers in effectively allocating funding and expediting the development of essential medical countermeasures. This approach ensures rigorously tested MCM and evidence-based decision-making. Finally, this report aims to assist the HERA in identifying areas for improvement in the strategic coordination of MCM development and enhancing the industrial capacity to produce and supply these countermeasures at the EU level.

1.5. AI-based tools to improve MCM development

AI is a rapidly evolving tool revolutionising many aspects of healthcare. It is a vast field that includes a variety of methodologies, such as computer vision, NLP, and ML. These methods are useful for pattern recognition, prediction, and large-scale data analysis.

In 2023, the Organisation for Economic Co-operation and Development (OECD) member countries approved a revised version of the organisation's definition of an AI system (OECD, 2023). The updated definition reflects developments over the last five years, improves technical accuracy, and makes it more 'future-proof'. The EU's forthcoming AI Act has aligned the definition of AI to the updated OECD definition in its final text. In this study, we will consider the updated OECD definition of 'AI systems' as a baseline to ensure technical accuracy and international alignment. The definition can be found in the box below (**Box *1***).

Box 1. Working definition of "AI system"

"An AI system is a machine-based system that, for explicit or implicit objectives, infers, from the input it receives, how to generate outputs such as predictions, content, recommendations, or decisions that can influence physical or virtual environments. Different AI systems vary in their levels of autonomy and adaptiveness after deployment."

Moreover, for this study, several AI techniques and algorithms have been considered. Particularly, the OECD Framework for the Classification of AI Systems provides a comprehensive and structured approach that helps characterise AI systems based on key dimensions (e.g. Data & Input, AI Model, and Task & Output). The framework provided by the OECD has been used as a guide to establish a classification model by providing a systematic way to assess and categorise different AI systems based on their characteristics and the way they interact with one another. Ultimately, the framework has allowed the study team to identify the potential risks and benefits associated with different AI systems, as well as inform how these AI techniques can be used for further development of policies and regulations related to AI and MCM development and supporting decision-making.
As a result, the study team considered three AI techniques. The table below provides an overview of the categories considered with their main characteristics and tasks performed (**Table *1***). On the one hand, inference using symbolic AI models allows for performing classification (e.g., Knowledge-based and expert systems). Statistical AI models facilitate trend and pattern prediction based on data (e.g., Linear regression, Logistic regression, Decision trees, Neural networks, Naïve Bayes, Deep learning). Lastly, hybrid AI models can combine previous approaches, to extract complex features, and enhance pattern recognition, and learning abilities.

Table 1.Definition of the AI techniques considered in this study

AI technique	Symbolic AI models	Statistical AI models	Hybrid AI models
Description	Symbolic or **knowledge-based** AI uses human-generated logical representations to infer a conclusion from a set of constraints (variables). These constraints include **rules**, ontologies and search algorithms and rely on explicit descriptions of variables and descriptions of the interrelations between these variables.	**Statistical AI models** (e.g. genetic algorithms, neural networks, clustering) identify patterns based on data rather than expert, human knowledge. They were previously used primarily for recognition, more recently, they are also being used for tasks like generation, such as synthesising and generating images or audio or identification of languages with large language models.	Many applied AI systems **combine symbolic and statistical** models into "hybrid" models. For example, NLP algorithms often combine statistical approaches building on large amounts of data and symbolic approaches that consider issues such as grammar rules.
Examples	Knowledge-based and expert systems	Linear regression, Logistical regression, Decision trees, Neural networks, Naïve Bayes, Deep learning	NLP
Data	Labelled and categorised	Unlabelled and uncategorised	Can be labelled or unlabelled
Task	Making inferences and performing classification	Trend/ pattern prediction	Complex feature extraction and pattern recognition/ learning abilities

Source: author's elaboration based on literature review findings.

Before 2019, AI has been predominantly employed in medicine and healthcare administration, however, the COVID-19 pandemic shifted its focus from medicine to public health (Olawade, 2023). AI advancements are expected to play a key role in MCM development which is an important component of health emergency preparedness and response (Chiu & Miller, 2019), helping to discern new patterns within molecular structures, identifying promising drug candidates and uncovering innovative solutions.

2. Results

The drug development process is traditionally made up of four broad stages according to the evidence (Ghandikota & Jegga 2024):

i. Identification of the target: protein target selection and validation
ii. Compound screening
iii. Pre-clinical studies
iv. Clinical trials

Recently, the US Food and Drug Administration (FDA) and the EMA recognised the increased use of AI techniques throughout the drug development life cycle. The number of drug and biologic application submissions to the FDA using AI or ML components over the past few years, with more than 100 submissions reported in 2021. These submissions include the different phases of drug development, from drug discovery and clinical research to post market safety surveillance (FDA 2024).

This section summarises the main results of the literature review and the enrichments of these with the information gathered in consultation activities with key experts in the area. It starts with the identified AI tools to support drug discovery, drug repurposing and drug target identification and continues with how AI techniques can support pre-clinical and clinical research as well as vaccine development.

2.1. AI for drug discovery and drug repurposing

The discovery and market introduction of a new drug can take up to 15 years, with an average cost of 2.6 billion US dollars and a high risk of failure (Ghandikota & Jegga 2024). Only 10% of compounds that enter phase I of clinical development proceed to the next phase, and the failure rate in phases II and III is around 50%, a figure that has remained unchanged for over two decades (Sun et al. 2022). Evidence suggests that approximately half of the failures are attributable to a lack of efficacy; often arising from a mismatch between preclinical and clinical results or the failure to meet predefined clinical endpoints. This underscores the challenges associated with identifying the appropriate druggable targets or biological mechanisms for the disease under investigation (Qureshi et al. 2023).

The global health crisis brought on by the COVID-19 pandemic highlighted the need for rapid, precise, and efficient drug discovery processes because of the unprecedented speed of the pandemic, the global demand for treatments and vaccines and the demonstrated potential for faster and more precise development approaches in identifying drug candidates. This highlighted the need for continued investment in innovative technologies such as AI, machine learning, and high-throughput screening to enhance efficiency and precision in drug development as well as encouraged ongoing advancements in technology, collaboration, and regulatory flexibility to better respond to future public health challenges. Traditional methods of drug discovery, which rely on in vitro high-throughput screening (HTS), demand substantial financial resources and sophisticated experimental infrastructures. However, AI tools hold the potential to transform the drug discovery process by enabling researchers to efficiently analyse large-scale data sets, design novel molecules, and predict the efficacy of potential drug candidates. Recent advancements have sparked considerable interest in applying AI across various stages of the drug discovery pipeline, including de novo molecular design, optimisation, structure-based drug design, and both preclinical and clinical development. These innovations offer a promising solution to reduce resource consumption while accelerating the identification of viable lead compounds (Choudhury et al. 2022).

The driving force behind using AI is its current capability to generate and access significantly more data than what is typically used in the drug discovery and development process. These datasets encompass both structured and unstructured information from a diverse range of public and private sources and are essential for training AI models, making high-quality and reliable data crucial (Arshadi et al. 2020).

Significant resources are being allocated toward the application of artificial inteligence in the discovery of novel molecules and medicines, given its vast and yet-to-be-fully-realized potential. In fact, according to the available evidence, it remains difficult to determine the effectiveness of these tools compared to traditional techniques (Prasad & Kumar 2021). In this regard, a recent report published by the Boston Consulting Group named *Unlocking the potential of AI in Drug Discovery*, examines the success rates of drugs discovered through AI in various phases of clinical trials. In Phase I, the study revealed that molecules discovered using AI have a success rate of 80-90%, significantly higher than the pharmaceutical industry's historical average of around 40-65%, however, the success rate decreases for Phases II and III (BCG). Besides that, questions arise from the use of AI in this context, especially when it comes to drug development during health crises and emergency response (Boston Consulting Group 2023).
In the next section, we explore in more detail the results of the use of AI tools to support drug discovery and drug repurposing, highlight successful strategies and tools and describe limitations and future developments.

2.1.1. Drug discovery

The main outcomes of the literature review on AI-based tools for drug discovery are:

- AI techniques such as ML, graphical neural networks or sequence-to-function frameworks can increase the efficiency and accuracy of the various stages of the drug discovery process, particularly by aiding researchers to target proteins or molecules that are crucial to disease.
- Generative models are an emerging tool in drug discovery, offering the ability to design novel drugs, protein therapeutics, and small compounds from scratch.
- AI can be used to identify potential new targets by analysing large amounts of data, identifying specific biomarkers or mutations that contribute to improve the understanding of disease mechanisms.
- Predictive modelling and virtual screening of molecule compounds help to prioritise the most promising candidates for further testing in the lab environment, saving time and resources.
- AI techniques can analyse vast biological datasets to identify crucial disease targets, enabling the development of more specific and effective drugs, which is vital during health emergencies like COVID-19 pandemic.

The main objectives of drug discovery are the identification of the **right target** in the organism, the design of the **right molecule** to interact with it and to determine in which patients the molecule is most likely to be used successfully (Hsieh et al. 2020).

The initial phase of discovering a new medicine involves **identifying the target**, which are usually genes or proteins believed to contribute to the development or aggravation of a disease (phase I). This is usually followed by *in silico* (computational) testing, performed on hundreds or thousands of chemical or biological compounds to evaluate their ability to specifically interact with the target molecules and the disease (phase II) (Diakou et al. 2022). After the discovery of the right compound, the pre-clinical research phase begins, where lead compounds are tested *in vitro* and *in vivo,* using experimental models that closely resemble human organism functions. Once these compounds are fully characterised, the most promising ones become lead candidates.

There are currently different types of AI technologies being used to support researchers in these activities as well as different approaches for using them. According to the evidence, all stages of the drug discovery pipeline could benefit from AI. For example, generative models have contributed to the design of new small molecules and protein therapeutics by generating novel drug-like compounds and to the development of protein therapeutics by creating de novo protein sequences optimized for specific

functions or properties; reinforcement learning can optimise molecular properties in specific directions and graph neural networks have been used to predict drug-disease associations. Additionally, NLP can mine scientific literature to find drugs and automate national agencies' approval steps (Qureshi et al.'s review, 2023). Overall, the rise of Machine Learning as analytical tools has led to an increase *in silico* methods, enabling the evaluation and organisation of large datasets from toxicology studies, as well as the modelling of drug toxicity and metabolism. Predictive *in silico* modelling is now applied in various fields, including industrial drug discovery, safety assessment of regulatory drugs, toxicity prediction of environmental agents, and food and cosmetic ingredient assessments (Jha et al. 2021). These methods are highly versatile, integrating expertise in computer science with deep knowledge of the specific scientific domains being explored (Jha et al. 2021).

For **the identification of the target**, AI algorithms have been used to analyse vast datasets of biological information, including protein structures and interactions, to identify potential targets within a disease or virus, such as specific proteins crucial for its survival. **By focusing on these targets, researchers can design drugs that more effectively interfere with the biological disease process, increasing the likelihood that the drugs will be effective by being specifically designed to alter the key mechanisms underlying the disease.** The role of AI for target identification is to analyse large amounts of biological data that include protein structure, protein-protein interaction between each other, information on genes and how their variation interacts with disease proteins to better understand their behaviour over time and their impact on the human body (Arshadi et al. 2020). A noteworthy example of such approach is the development of AlphaFold and RoseTTa Fold, which allowed the systematic fold exploration of the proteome by generating the structure of proteins that had not been yet elucidated (Jumper et al. 2021; Baek et al. 2021). AI enables the modelling of proteins without previously available structures, facilitating structure-based drug development for targets that were previously inaccessible. A **well-chosen target allows for the development of drugs that act more specifically and selectively on the disease**, and this is key, especially for drugs developed during health emergencies, for minimising side effects and increasing the effectiveness of the treatments (Diakou et al. 2022).
In the context of health emergencies and response, AI-assisted drug discovery offers significant promise for addressing challenges in infectious diseases but faces hurdles due to imbalanced datasets and the unique complexities of pathogens. Pathogen-specific targets often suffer from underrepresentation in training data, especially for neglected diseases, and the rapid evolution of resistance poses additional difficulties. Despite these challenges, AI has demonstrated success in identifying novel antibiotics, such as *halicin*, and accelerating antiviral and vaccine development, particularly during the COVID-19 pandemic (Booq et al. 2021). Advances in generative models, transfer learning, and structural biology tools like AlphaFold enable AI to design innovative compounds and predict druggable targets for pathogens. However, the clinical translation of AI-predicted candidates remains a bottleneck, highlighting the need for better datasets, cross-disciplinary efforts, and real-time adaptability to emerging infectious threats.

Section 2.2 shows in more detail the AI techniques and methods that have led today to the identification of the targets. Evidence describes that the main techniques used for target identification are ML models, including graph neural networks, SE(3) Transformers, sequence-to-function and sequence-to-structure frameworks including examples such as AlphaFold, and generative models (Wong et al. 2023).

For compound screening, Virtual Screening (VS) is a widely employed computational technique that identifies molecules capable of binding with proteins or enzymes, utilising predictive modelling for drug development. **By simulating how drugs might interact with biological systems, VS enables the evaluation of millions of compounds, allowing researchers to virtually screen large chemical libraries to identify those most likely to be effective and safe** (Ahmed et al., 2022). This approach significantly accelerates the initial stages of drug discovery by narrowing down potential candidates for further in vitro and in vivo testing (Ahmed et al. 2022).

In addition, AI plays a pivotal role in enhancing the screening of molecules for intracellular pathogens by addressing critical challenges such as target accessibility, host-pathogen interactions, and the scarcity of pathogen-specific data. AI can design molecules optimised for intracellular delivery, increasing their likelihood of effectively reaching and interacting with pathogen targets within host cells. Moreover, AI models are adept at predicting the evolution of resistance, helping to identify drug candidates with a higher barrier to resistance development. By utilising multi-omics data and advanced machine learning algorithms, AI improves the specificity, efficiency, and overall success rate of drug discovery efforts for complex infections, ultimately reducing the time and costs associated with traditional drug development methods. The integration of AI-driven insights can thus streamline the identification of promising therapeutic candidates for intracellular pathogens, which are often difficult to target with conventional approaches (Wong et al. 2023).

The growing availability of extensive chemical libraries also supports docking simulations, through which researchers can determine which drug candidates from the library bind effectively to target proteins. One of the most promising tools in this domain is AlphaFold3, which has revolutionized the ability to predict protein structures and their interactions with small molecules with unprecedented accuracy (Abramson et al., 2024). **The synergy of predictive modelling and Virtual Screening enables the prioritization of the most promising candidates, allowing researchers to focus on the most likely effective compounds, thus optimizing laboratory testing and conserving valuable time and resources** (Park et al., 2022).

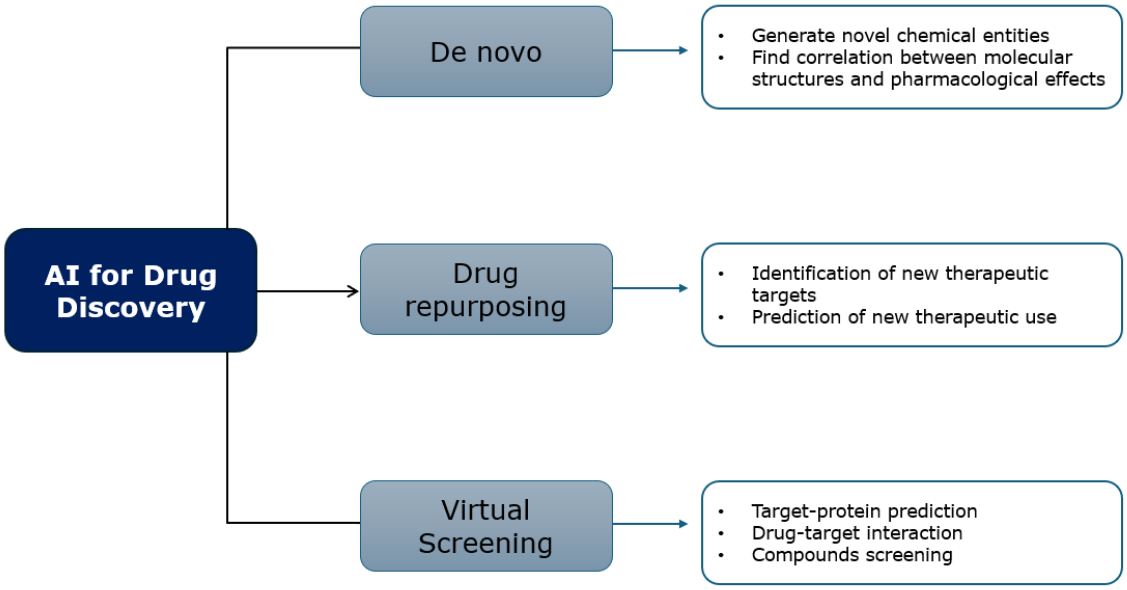
According to the evidence, the generic definitions of VS and predictive modelling are significantly broader and may encompass many different methods. An explanation of these methods and approaches is described in section 2.2. This analysis is complemented by a careful selection of VS success case studies (Panwar et al.'s review).

The use of generative models for designing *de novo* drugs, small molecules, and protein therapeutics is an increasingly important and emerging field in drug discovery. These models allow the rational design of drugs from scratch, tailored for specific applications or targets. As protein-based therapeutics become more widely investigated in preclinical and clinical research, optimising their developability has become a resource-intensive and time-consuming process. This has led to a growing need for strategies that can predict and enhance the developability of these therapeutics at an early stage (Chen et al. 2023).

Given the vast search space and complexity of protein drugs, advanced approaches are required to streamline their design and optimise their properties. Advanced computational methods, which have rapidly evolved, offer strong capabilities in feature extraction and modelling, enabling a deeper understanding of the complex rules governing protein properties. These methods provide a multi-scale perspective that has the potential to significantly accelerate the development of protein therapeutics while reducing the associated costs compared to traditional experimental approaches (Samidh 2017).

Based on the predicted interactions with identified targets, AI can suggest existing drugs that might be repurposed for the new disease or propose entirely new drug candidates. The "**repurposing**" approach can significantly reduce the time and resources needed compared to traditional drug discovery methods, which often start from scratch, especially because these kinds of drugs have already undergone safety control protocols (Gupta et al. 2023). This concept is reviewed in the following section.

Figure 1. A conceptual overview of the use of AI for drug development.



Source: Based on the literature review study.

2.1.2. Drug repurposing

The main outcomes of the literature review on AI-based tools for drug repurposing are:

- Drug repurposing involves advanced computational algorithms to predict structures of drug targets, detect interaction hotspots, and screen known drug candidates against new target structures. This approach can expedite the discovery of therapeutic uses for existing drugs, offering several advantages such as reduced development times and lower risk of failure due to existing safety profiles.
- AI and computational methods allow the analysis of large-scale datasets to predict drug-target interactions and enable more efficient and accurate predictions.
- AI techniques have played a pivotal role in enabling researchers to accurately determine the protein structures of viruses. The understanding of viral proteins has allowed the identification of the critical virus mechanisms, facilitating the design of drugs or vaccines that target these essential components more accurately and effectively.
- Despite the promising potential of data and computational science, challenges such as limited data availability and quality, clinical trial bottlenecks, and variability in human genetic responses that impact disease severity and treatment efficacy have been reported when implementing drug repurposing.

This section discusses AI-based computational methods and tools applied to drug repositioning - commonly called “repurposing” or “reprofiling” - in which drugs with previously known indications are repurposed for new indications (Dotolo et al. 2021 Aronskyy et al. 2021; Prasad & Kumar 2021; Choudhry et al. 2022; Zong 2022: Singh et. Al 2023; Sundari et al. 2024; Ghandikota & Jegga 2024). This field acquired considerable importance during the COVID-19 pandemic, in which drug repurposing played an important role in reducing costs and times for the marketing of medicines useful for dealing with emergencies (Dotolo et al. 2021).

Drug repurposing provides a promising alternative to finding new therapeutic uses for existing drugs that were previously approved for other indications reducing the pre-clinical phases of drug development (Choudhury et al. 2022). In this strategy, advanced computational algorithms can predict 3D structures of targets, detect binding pockets/interaction hotspots of new drug targets, and screen the already

known drug candidates against new target structures (Choudhury et al. 2022). Drug repurposing offers several advantages compared to developing an entirely new drug. Firstly, the risk of failure is lower because the repositioned drug has already been found sufficiently safe in pre-clinical animal models and human models. Secondly, drug development times can be reduced since most pre-clinical tests related to safety evaluation and, in some cases, formulation development, have already been completed (Sundari et al. 2024).

According to the literature, during health crises and emergencies, following a drug repurposing approach allows for a timelier action. For example, during the COVID-19 pandemic, advances were reached by leveraging the wide safety profiles of existing drugs and quickly screening them for potential antiviral activity. For instance, *baricitinib*, an anti-inflammatory drug approved for rheumatoid arthritis, was repurposed and authorised for emergency COVID-19 treatment based on encouraging trial results. The authors concluded that such drug repositioning efforts hold potential for the timely development of MCM against future novel infectious disease outbreaks (Ghandikota & Jegga 2024).

The evidence highlights that the success of drug repurposing has been established in recent years thanks to the increasing production and collection of data. In fact, in the last decade, the rapid accumulation of diverse chemical, genomic, proteomic, and clinical datasets, has produced a large-scale domain of critical data that only computational methods can explore properly (Ghandikota & Jegga 2024). Some examples are an AI model identifying existing antiviral drugs such as remdesivir and chloroquine as potential candidates for COVID-19 treatment. While some of these repurposed drugs were eventually found to be less effective in clinical trials, the rapid computational screening helped to accelerate the search for viable options (Sing 2024). Another example is the use of the SARS-CoV-2 virus genomic sequence to predict viral proteins to be used as targets for vaccines through AI models (Choi et al. 2024). To process these data, usually available in digital libraries, deep learning tools are proven to be effective given the ability of AI techniques to learn complex relationships between input features and output decisions from large-scale data, supported by the continuous increase of computing power from hardware equipment. (Pan et al. 2022). Pan et al. (2022) also noted that recent advances in deep learning and the availability of public databases have boosted the research in drug repurposing. For instance, computational methods, especially molecular docking, are used to identify interactions between drugs and targets, employing various AI-based strategies and tools (Prasad and Kumar 2021). The **drug repurposing workflow** is organised differently from traditional drug development and includes fewer steps and different parameters to follow. It capitalises on existing drugs, which have already undergone substantial research and development, including safety and efficacy evaluations (Dotolo et al. (2021), Aronskyy et al. (2021):

i. Compound identification/selection: For drug repurposing, this step is bypassed because the compounds are already established drugs. Instead of identifying new compounds, the focus is on selecting existing drugs that may be effective for new therapeutic indications;
ii. Compound acquisition: In drug repurposing, this phase is simplified since the drugs are already available and can be sourced from well-known chemical databases and libraries;
iii. Development: Since the drug molecules are already approved and safety well-established, the preclinical studies are often bypassed;
iv. FDA post-market safety monitoring: the safety of the drug is documented in previous safety monitoring, so this phase is simplified and faster.

Various computational and AI-based approaches for drug repurposing have been reported in the literature. Dotolo et al. (2021) and Aronskyy et al. (2021) categorize these approaches into four groups:

- molecular docking
- network-based models
- structure-based approaches

- AI based approaches and hybrid approaches.

As molecular docking is frequently incorporated into network-based models, we address them together below:

Network Based approach

Network-based approaches play a vital role in drug repurposing due to their ability to integrate multiple data sources. With advancements in high-throughput technology and bioinformatics, molecular interactions within biological systems can be modelled as networks. In these networks, nodes represent drugs, diseases, or gene products, while edges denote interactions or relationships. This type of modelling aids in structure-guided pharmaceutical and diagnostic research by pinpointing potential new biological targets (Dotolo et al. 2021).

Research has shown that networks involving drug–target, drug-drug, drug-disease, and protein interactions are effective in uncovering new opportunities for drug discovery and repositioning (Dotolo et al. 2021). Network-based methodologies use a systems pharmacology-based network medicine platform, which quantifies the interactions between virus-host and drug targets within the human protein-protein interaction network (Choudhury et al. 2022).
An example of the use of network-based approaches was conducted during the COVID-19 pandemic, as reported by Dotolo et al. (2021), where two main types of network-based approaches were prevalent:

- Clustering approaches: clustering approaches discovered novel drug-disease or drug–target relationships by finding modules (such as drug-disease, drug-drug, or drug–target) using clustering algorithms based on network topology (Dotolo et al. 2021, Choudhury et al. 2022).
- Propagation approaches: in the COVID-19 pandemic, propagation approaches involved collecting human coronavirus-associated host proteins from literature and creating a pan-HCoV protein subnetwork. The proximity between drug targets and HCoV-associated proteins was calculated to identify candidate repurposable drugs for HCoVs within the human protein interactome model. These methods helped to annotate proteins functionally associated with HCoVs and model proteins that serve as drug targets for specific diseases, potentially identifying suitable drug targets for antiviral infections (Dotolo et al. 2021, Choudhury et al. 2022).

Network-based approaches for drug repurposing have proven highly effective across a wide range of diseases beyond COVID-19, leveraging biological networks to identify potential therapeutic uses for existing drugs. By analysing complex interactions between genes, proteins, and disease pathways, these approaches can uncover novel drug-disease connections. For example, kinase inhibitors originally developed for cancer have shown promise for Alzheimer's disease, while antidepressants like fluoxetine have been repurposed for Parkinson's disease. In oncology, antiviral drugs like oseltamivir have demonstrated efficacy in treating glioblastoma, and ethambutol, an anti-TB drug, has the potential to treat Crohn's disease. Similarly, famotidine, a histamine receptor antagonist, is being investigated for cancer therapy. These examples highlight how network pharmacology and gene-drug interaction networks provide valuable insights for repurposing existing drugs, and speeding up the discovery of treatments for diverse and complex diseases. However, Network-based drug repurposing for infectious diseases faces several challenges due to the relatively limited availability of comprehensive data compared to diseases like cancer or neurodegenerative disorders. The complexity of infectious diseases, which often involve rapidly evolving pathogens, makes it difficult to establish robust biological networks. Pathogen-specific data, particularly for diseases caused by viruses, bacteria, or fungi, is less abundant, and the rapid mutation rates of pathogens like SARS-CoV-2 or antibiotic-resistant bacteria can outpace the development of network models. Furthermore, clinical data for infectious diseases, especially emerging ones, is often sparse, and preclinical data may not always translate effectively to human outcomes. Additionally, there are concerns about antimicrobial resistance and the regulatory hurdles associated with repurposing drugs for infections. Despite these limitations, advances in genomic

and proteomic technologies, as well as the growing availability of pathogen-specific datasets, offer potential for overcoming these challenges in the future.

Overall, network-based methods provide a general overview, because they can analyse biological networks to map interactions between the virus, host proteins, and cellular pathways. By identifying drugs that target these interactions and potentially disrupt the virus's life cycle, researchers can uncover promising candidates. In addition, network-based approaches can be used even when there is limited data on the interactions between the virus and human proteins (Aronskyy et al. 2021).

Structure-based approaches

Structure-based approaches for drug repurposing leverage knowledge of the three-dimensional structures of biological targets, such as proteins or enzymes involved in diseases, to identify new therapeutic uses for existing drugs (Choudhury et al. 2022). This kind of approach is often based on molecular docking, a technique that virtually simulates how a drug molecule fits into the binding pocket of a viral protein. **By analysing the fit and interactions, scientists can predict if the drug can inhibit the protein's function and potentially stop the virus in its tracks** (Aronskyy et al. 2021).

Advances in ML, particularly deep-learning techniques, are increasingly being integrated with structure-based methods like molecular docking to enhance the prediction of drug-target interactions. According to the available literature, the growth in drug-related datasets and open data initiatives has facilitated the development of sophisticated ML models that can use simplified vector representations of molecular structures, such as molecular fingerprints, instead of relying solely on detailed 3D structures. ML models are trained on datasets that pair molecular descriptors with binding affinity data, enabling them to predict binding affinities efficiently even in the absence of a solved 3D protein structure. This integration not only enhances computational efficiency but also broadens the scope of potential drug candidates (Dotolo et al. 2021). For instance, studies have demonstrated the effectiveness of ML models in predicting high-affinity drug candidates for SARS-CoV-2 targets and in identifying promising drugs that might be repurposed for COVID-19 treatment. A notable finding is the work by Gao et al.(2020), where structure-based drug repurposing combined with ML was employed to identify inhibitors of the SARS-CoV-2 main protease. In that case, a gradient-boosting decision tree model was built using 2D fingerprint data to represent the drugs, then the decision tree model was trained with data from known inhibitors to better understand how the structure of the molecule behaves to inhibit the main SARS-CoV-2 protease 3CLpro (Prasad and Kumar report). Another example is the use of the structure-based drug repurposing approach to predict the binding affinity of the SARS-CoV-2 viral spike protein and the spike-human ACE2 interface (Batra et al. 2020). They trained a structure-based ML model to repurpose drug candidates against the spike protein and the spike–human ACE2 interface complex. The model used as input a set of features that capture the interactions between the drug molecule and the target proteins at various spatial scales. As output, the model predicted the corresponding AutoDock Vina score, a widely used 3D structure-based docking algorithm. Predictions were based on a random forest model trained on docking-generated binding affinities. Top-scoring ligands, including FDA-approved antivirals and antibacterials, were further validated *in silico* by all-atom docking studies (Aronskyy et al. 2021).
In general, by combining the detailed structural insights from molecular docking with the computational power and flexibility of ML, researchers achieved more accurate and comprehensive predictions, accelerating drug discovery and repurposing processes (Dotolo et al. 2021).

AI-based and hybrid approaches

Recent studies on drug repurposing are based on research conducted during the COVID-19 pandemic since considerable funding was allocated during that period to drug repurposing. Most of the developed applications were based on ML that contributed substantially to the repurposing of old medicines, benefiting from the *de novo* approach. AI techniques have aided researchers in determining the protein

structures of the virus, allowing to identify the necessary machinery for designing a drug or vaccine more accurately and effectively (Prasad and Kumar 2021). Scientists from the AI drug discovery company BenevolentAI and Imperial College London used their in-house developed algorithms to mine data and identify the enzyme adaptor-associated protein kinase 1 (AAK1), as a possible target for COVID-19. The program then reported the above-mentioned drug, *Baricitinib*, an approved drug against rheumatoid arthritis, as one of the best inhibitors out of 378 known AAK1 inhibitors (Richardson et al., 2020).

Other examples are the use of used NLP to develop a tool called BERE to identify drug and virus interactions in scientific texts (Ge et al. 202); Beck et al. (2020) applied deep learning with the Molecule Transformer-Drug Target Interaction (MT-DTI) to predict drug interactions with SARS-CoV-2 proteins; Zhang et al. (2020) used a large knowledge graph, CoVKGE, to find drug-disease relationships; Esmail and Danter (2020) employed a hybrid approach with DeepNEU v5.0 to simulate lung cells and test existing drugs; Ahmed et al. (2022) developed SperoPredictor R to predict drug interactions with disease proteins and validated these with literature and simulations and Matu Kumar et al. (2021) used AI for VS and simulations to identify promising drugs targeting SARS-CoV-2's main protease. Detailed information on these methods is summarized in **Table 2**.

Various network medicine platforms and computational tools were developed to identify repurposable drugs for COVID-19. These platforms analysed virus-host interactions and drug targets to prioritise candidates through AI techniques such as network diffusion to predict and validate effective drugs, highlighting the importance of integrating different techniques for drug discovery. For example, the combination between *baricitinib* and *ritonavir* therapies showed positive results. The studies suggest that *baricitinib* could be shown to be an effective and safe alternative in selected hospitalised patients with SARS-CoV-2. More information and clinical trial results are needed to understand which patients could benefit from treatment with cytokine inhibitors and the optimal timing of their administration. Tools like the COVID-19 Disease Map provide open-access resources for understanding molecular mechanisms and supporting drug development (Turilli et al. 2022). Open-access resources are important tools for AI-driven drug discovery and repurposing, as they provide the vast and diverse datasets needed to train and validate predictive models. By offering free and transparent access to databases such as PubChem, DrugBank, and ChEMBL, researchers can explore molecular structures, pharmacological data, and clinical trial outcomes without prohibitive costs. This fosters global collaboration, accelerates innovation, and enables cost-effective research, particularly in low-resource settings. Open-access data also play a crucial role in drug repurposing, allowing AI to identify new therapeutic applications for existing drugs by mining openly available clinical and omics datasets. Moreover, during public health emergencies like the COVID-19 pandemic, open-access initiatives such as CORD-19 have shown their value in rapidly facilitating discoveries. While challenges like data standardisation and privacy concerns remain, open-access resources are indispensable for equitable and impactful advancements in AI-facilitated drug research.

Hsieh et al. (2021) created a comprehensive COVID-19 knowledge graph including interactions between SARS-CoV-2, host genes, pathways, drugs, and phenotypes using Graph Neural Networks (GNN) via link prediction, achieving a high level of accuracy. In addition, Li et al. (2021) manually compiled a dataset of experimentally confirmed virus-drug associations, including 175 drugs, 95 viruses, and 933 virus-drug interactions from publicly available databases and literature. Due to the sparsity and imbalance of the data, they developed an ensemble model called EMC-Voting, which combined matrix completion and weighted soft voting in a semi-supervised ML framework. The model was evaluated using fivefold cross-validation and compared against other classifiers and prediction models. The case study for SARS-CoV-2 showed that EMC-Voting achieved a high area under precision-recall curve value of 0.934, indicating strong performance in predicting virus-drug associations (Li et al. 2021). Additional advances in this field are the Exscalate4CoV project, a European public-private consortium, conducted extensive virtual screening and drug repositioning experiments, identifying raloxifene as a potential

treatment for early-stage COVID-19 patients[1]. Clinical trials were authorised in October 2020 to evaluate its efficacy in preventing disease progression. Similarly, olgotrelvir (STI-1558), an experimental antiviral targeting the SARS-CoV-2 main protease, has shown promising progress (Mao et al. 2023).

2.1.3. Importance of critical data in Drug discovery and repurposing

Based on the available evidence, it is essential to generate both absolute and methodological data in drug discovery to address numerous challenges associated with the validation of ML techniques. These challenges include enhancing decision-making processes, raising awareness of ML methodologies, and identifying existing gaps and challenges (Dara et al., 2022). Since ML techniques use mathematical algorithms to develop models from datasets, the performance of these models can improve over time by incorporating various types of data (Iqbal et al., 2023, p.3).

Overall, the articles reviewed in this report suggest that the efficacy of AI in drug discovery and repurposing is contingent upon the availability of critical data for processing. Das et al. (2021) provided an in-depth analysis of the AI strategies for addressing emergencies, such as COVID-19, by employing data science expertise to select appropriate tools. Their review categories COVID-19 data into several critical sources: Omics Data, which includes genomics for tracking viral mutations and origins, proteomics for studying virus-cell interactions, and metabolomics for identifying biomarkers; Clinical Data, which encompasses electronic health records, test results, and imaging data such as X-rays and CT scans essential for understanding disease progression and treatment responses; and Epidemiological Data, which includes case numbers, mortality rates, geographic spread, and transmission factors crucial for outbreak management. The authors highlighted prominent public repositories, such as GISAID for viral genomics, the Johns Hopkins University COVID-19 dashboard for epidemiological data, and the Gene Expression Omnibus for gene expression data, which facilitate data sharing among researchers (Clough & Berret 2016, GISAID 2024, Johns Hopkins 2024). They also address the challenges of data integration due to inconsistencies in data formats and collection methodologies, stressing the need for standardised practices. The article explores various data science techniques used to analyse COVID-19 data, including ML for drug repurposing, network analysis for understanding viral interactions, image analysis for diagnostic automation, and predictive modelling for outbreak control. Despite the promising potential of data science, the authors acknowledge challenges such as limited data availability, data quality issues, clinical trial bottlenecks, and variability in human genetic responses that impact disease severity and treatment efficacy. Genetic background plays a critical role in AI-driven drug repurposing by influencing how individuals respond to treatments. Genetic variability can affect drug metabolism, receptor binding, and the activation of targeted pathways, making it essential to consider genetic factors when predicting drug efficacy. AI models can integrate genomic data to identify genetic markers associated with diseases or drug responses, enabling precision medicine and enhancing the repurposing of drugs for specific genetic subgroups. By creating gene-drug interaction models and incorporating pharmacogenomic insights, AI can optimize treatment recommendations, ensuring more personalized, effective therapies and increasing the likelihood of success in diverse patient populations.

In this context, both the FDA and the EMA are dedicated to ensuring that medicines are both safe and effective, while also fostering innovation in drug development. As with any emerging technology, AI and ML present both opportunities and novel challenges. To address these challenges, the FDA has accelerated its efforts to establish a responsive regulatory framework that supports innovation while protecting public health. Specifically, the FDA's Center for Drug Evaluation and Research, in conjunction with the Center for Biologics Evaluation and Research and the Center for Devices and Radiological Health, has issued an initial discussion paper to engage with stakeholders and examine pertinent

[1] To have more information please visit: Exscalate4CoV, from https://www.exscalate4cov.eu.

considerations regarding the application of AI/ML in drug and biological product development (FDA 2023). The FDA intends to create and implement a flexible, risk-based regulatory framework that balances innovation with patient safety.

Similarly, the EMA released a draft paper in 2023 addressing the use of AI throughout the medicinal product lifecycle. The EMA emphasized the need for integrating data science expertise across all phases of medicine development and pharmacovigilance. The agency underscored the importance of prioritizing patient safety and maintaining the integrity of clinical study results by mitigating the risks associated with new AI technologies and called for measures to prevent biases in AI applications and enhance the trustworthiness of AI systems (EMA, 2023).

The number of available public and private chemical libraries is vast and continually expanding, including an extensive array of compounds for drug discovery and development; the review by Choudhury et al. (2022) provides a well-detailed list of data sources for repurposable chemical space, targets, pathways, and drug–target complexes using ML. Also, the work of Ahmed et al. (2022) can be used as a reference, providing a detailed scenario on the importance of critical data for drug development, and, particularly, on the creation of a dataset and merging of different data sources using cross-database identifiers (e.g PubChem, SIDER, Uniprot, ChEMBL, etc.). Lastly, the work of Wang et al. (2021) underscores the critical importance of access to high-quality data for training AI models, particularly for deep learning applications, stressing the need to develop robust models capable of predicting new antiviral drugs for pathogen outbreaks and pandemics (Wang et al. 2021).

Box 2. Considerations from the experts: AI for drug discovery and drug repurposing

Highlights resulting from Experts' consultation

According to experts from the Academia, AI-based software is becoming increasingly critical in drug development, particularly for the accurate prediction and design of 3D protein structures and their interactions with other proteins or ligands. High-quality data for training these AI models is becoming ever more important, and future advancements will likely depend on new, untapped data sources. This could include synthetic datasets or active learning approaches. However, according to them, there are limits to existing resources, such as the **Protein Data Bank**, which contains finite information for training AI models. As the data from this resource is exhausted, access to proprietary structural biology data, held by large pharmaceutical companies, will become crucial. Although most of this data remains inaccessible due to its proprietary nature, strategic partnerships, like those between **Isomorphic Labs**, **Eli Lilly**, and **Novartis**, offer potential solutions by granting access to valuable x-ray data of small molecules and protein targets. In the coming years, access to data will become one of the most significant challenges in AI-driven drug development, though synthetic data and active learning offer promising alternatives to mitigate this challenge.

An expert with both clinical and research expertise noted that the leading research institutes in drug discovery are currently building protein design models similar to **AlphaFold** but incorporating generative models like **RFDiffusion** and **RF Antibody Diffusion**, which are already capable of designing novel protein structures. She emphasised that models such as AlphaFold, **RoseTTAFold**, and **ESMFold** can predict protein structures even when no crystal structures are available While these models predict protein structures primarily from sequence data, understanding protein interactions with small molecules can complement their predictions by providing insights into functional conformations or binding dynamics.

In the expert's opinion, AlphaFold's structure prediction models, particularly AlphaFold3, are useful for small molecule docking but have not yet proven effective for small molecule design - an area where AI could provide significant advancements. She added that companies like **Recursion** and

other AI-driven small molecule discovery firms could play a major role in informing synthesis pathways for manufacturing, production, and novel, more efficient synthesis methods. She concluded that the immediate application of AI in protein design is more advanced compared to its application in small molecule design. Academic experts agreed, highlighting the remarkable potential of generative protein design tools to create *de novo* proteins capable of interacting with other proteins or DNA and possibly, in the near future, with enzymes enas well. Unlike structural or binding proteins, enzymes must achieve precise conformations and maintain dynamic flexibility to facilitate chemical reactions efficiently. This complexity makes it more difficult to predict how changes in enzyme structure affect function, which is critical for tasks such as small molecule docking or design. While AlphaFold and its iterations, such as AlphaFold3, have shown remarkable accuracy in structure prediction, their utility in small molecule design remains limited. Designing small molecules requires predicting not just how a molecule interacts with an enzyme but also how to optimise that interaction for specificity and efficiency, tasks that AI models are still developing. In this regard, targeting enzymes like proteases and polymerases is crucial in infectious disease drug development, but it presents challenges. Enzymes are dynamic, making it difficult to predict their active states and design effective inhibitors. Additionally, mutations in microbial enzymes can lead to drug resistance, and achieving specificity to avoid off-target effects is challenging. Limited structural data further complicates drug design. While AI tools like AlphaFold have advanced static structure prediction, their application to dynamic enzyme modelling and small molecule design is still evolving. However, AI holds promise for overcoming these obstacles through dynamic modelling, resistance prediction, and novel inhibitor design.

Another point mentioned during the interviews by industry experts was that AI can be used to model the stability of a drug and predict its shelf life, offering significant advantages over traditional methods. They mentioned that various approaches have been developed, including computational modelling of factors like the degradation profile or environmental "shadow" of a product, which is crucial for determining its shelf life. For newly launched or repurposed drugs, regulatory approval often depends on these stability studies, which typically require 12 months of data. Computational modeling has become an essential tool for determining the shelf life of products by analysing key factors like degradation profiles and environmental influences. These models simulate how products deteriorate over time under varying conditions, such as temperature, humidity, and light exposure, providing insights into their stability and longevity. For instance, degradation profile modeling predicts the rate at which active ingredients lose efficacy, while environmental "shadow" analysis assesses the impact of external factors like oxygen levels and microbial contamination. Such predictive approaches are widely used across industries, from pharmaceuticals and food to cosmetics and materials, enabling manufacturers to optimise packaging, storage, and transportation strategies. By offering faster and more accurate shelf-life predictions compared to traditional methods, computational modeling reduces waste, enhances quality control, and supports sustainable practices.However, AI-driven shelf-life prediction models could potentially shorten this time frame by making accurate predictions based on just three to six months of data, significantly reducing the time and resources needed for stability testing.

The table below provides an overview of the main AI-based tools identified for drug discovery and repurposing in health emergency preparedness and response, the AI technique utilised, and their implementation in the market (**Table 2**).

Table 2. List of AI tools identified for drug discovery and repurposing

AI-based tool	AI Techniques	Functioning and goals	AI-based tool author	Availability[a]
AutoDock Vina	ML	Bind affinities generated by docking simulations	Batra et al. 2020, quoted in	Open

			Aronskyy et al. 2021	
Benevolent AI	Knowledge graph	Mine the data and find the enzyme, adaptor-associated protein kinase 1 (AAK1) as a possible target for the COVID-19	Richardson et al. 2020	Public
BERE	NLP	Identify drug and virus interactions in scientific texts	Ge et al. 2020	In-house
Molecule Transformer-Drug Target Interaction (MT-DTI)	Deep Learning	Predict drug–target interactions	Beck et al. 2020 quoted in Aronskyy et al. 2021	In-house
CoVKGE	Knowledge graph	To find drug-disease relationships	Zhang et al. 2020	In-house
DeepNEU v5.0	Deep Learning	Simulate lung cells and test existing drugs	Esmail & Danter 2020	In-house
SperoPredictor R	ML	Predict drug interactions with disease proteins and validate evidence with literature and simulations	Ahmed et al. 2022	In-house
AutoQSAR	ML	Predict the antiviral activities of compounds	Kumar et al. 2021	Public
QikProp	ML	Examine molecules for their drug-likeness and toxicity parameters	Kumar et al. 2021	Open
Prime-MM/GBSA algorithm	ML	Examine the binding free energy of molecules against the main protease of Covid-19	Dasmahapatra et al. 2022	Public
AlphaFold	Deep Learning	Predicts a protein's 3D structure from its amino acid sequence	Jumper J et al. 2021	Open
RosetTTAFOLD	Deep Learning	Generate accurate models of protein-protein complexes	Baek M et al. 2021	Open
RFDiffusion	Diffusion model (ML)	Enables the design of diverse functional proteins from simple molecular specifications	Watson J et al. 2023	Open
ProteinMPNN	Deep Learning	Folding model to predict amino acids from a protein structure and certain chains and complexes	Dauparas J et al. 2022	Open
Papyrus	ML	Dataset that contains multiple large publicly available datasets such as ChEMBL and ExCAPE-DB combined with several smaller datasets containing high-quality data	Bequignon O et al. 2023	Open

[a] *For this study the availability of the tool in the market was defined as: "Research use" for AI-based tools developed within the academia but not yet implemented or tested within a real-world setting, "In development" for AI-based tools currently under design or improvement. Overall, the methodology of these tools has been published in the literature because of their innovative approaches, "Open source" for AI-based tools for which the original source code is made freely available and can be redistributed and modified, and "On the market" for AI-based tools that have been already deployed in real-world settings and are currently being used.*

2.2. AI for drug target identification (DTI)

The main outcomes of the literature review on AI-based tools for DTI are:

- AI systems are capable of analysing various types of data, including genetic, proteomic, and clinical information, to pinpoint potential therapeutic targets. By revealing disease-related

targets and molecular pathways, AI plays a key role in developing drugs that can influence biological processes.
- Computational strategies and ML play a crucial role in DTI by identifying unique compounds within large databases. Deep learning technology is essential for accurately predicting a drug's affinity for specific targets, expediting the medication formulation process and offering superior performance compared to other methods.
- ML models were developed to screen large libraries of molecules for their effectiveness against SARS-CoV-2, demonstrating the potential of these technologies to accelerate identification of new compounds.
- The comparison of chemical compounds and predictive modelling through ML techniques significantly accelerates DTI discovery by leveraging computational power to prioritize the most promising candidates for experimental validation.

The drug-target identification (DTI) is a highly complex process that integrates various scientific fields, such as computational chemistry, medicinal chemistry, and chemoinformatics, providing a comprehensive approach to the process (Huang et al. 2020). The design of novel small molecules based on their properties has gained significant attention nowadays, with billions of molecules now being available in public and commercial databases (Huang et al. 2020). **Identifying unique compounds within these large databases is a crucial and complex task since computational strategies**, including AI techniques and ML algorithms, are frequently employed to find suitable compounds (Huang et al. 2020). For instance, deep-learning technology, which employs neural networks, is crucial for accurately assessing and predicting a drug's affinity for specific targets. It can save time and economic resources by expediting the medication formulation process with discretion and deliberation. Due to its superior performance compared to other methods utilised in the industry, deep learning is now widely accepted as the most important method for predicting bioactivity (Sayal et al. 2024, p. 191).

Generally, during DTI, ML or deep-learning techniques are used to analyse public interactive databases, such as repositories in which discovered molecules are already included. The importance of these libraries is discussed by several authors and researchers, who have tested their usefulness (Wang et al. 2021). Several studies included in the present report highlight the potential of ML or deep learning models in accelerating DTI for emerging infectious diseases.

For example, in the study by Wang et al. (2021), authors aimed to identify potential drug candidates for treating COVID-19 by developing a deep learning model called COVIDVS. This model was designed to screen large libraries of molecules and predict their effectiveness against SARS-CoV-2. The researchers compiled data on molecules known to be effective against various coronaviruses, including SARS-CoV and MERS-CoV, incorporating information on molecular structures and antiviral activities. Molecules with documented antiviral properties were labelled as "positive" data points, while those without such properties were labelled as "negative". To analyse these molecules, researchers employed a deep learning technique known as a directed message-passing neural network (DMPNN), which examines molecular structures represented as graphs and identifies patterns linked to antiviral properties. The DMPNN model was trained on the compiled antiviral data, allowing it to learn critical structure-activity relationships. Once fine-tuned, COVIDVS was used to screen extensive libraries of drug-like molecules, compounds in clinical trials, and pre-clinical tool compounds. For each molecule, the model predicted a score indicating its potential effectiveness against SARS-CoV-2. In addition, ML models developed for SARS-CoV-2 screening are highly adaptable to other viruses and pathogens. By leveraging large datasets, these models can identify potential inhibitors for similar viruses, like MERS-CoV or influenza, and even bacteria, fungi, and parasites. However, the success of ML in this area depends on the availability of quality data specific to each pathogen and its unique biological mechanisms. While ML can accelerate drug discovery across various diseases, each new pathogen

may require tailored models to account for differences in molecular targets and resistance patterns. As data grows, ML's potential in broad pathogen screening continues to expand.

The review by Gupta et al. (2023) and El-Behery et al. (2021) addresses the techniques for elucidation of DTI and efficacy, which are crucial steps in developing new drugs. According to the authors, ML has been used to help screen drug targets and design peptides. In addition, the authors mention that ML has been used to mine bioinformatics data and analyse biological pathways to identify novel pathways that can lead to a greater understanding of the disease mechanism (Auwul et al. 2021). AI has also assisted in the analysis of samples to support rapid diagnoses with a less expensive assay that is highly sensitive, selective, and accurate (Jaroenram et al., 2022; Lai et al., 2022).

In addition, Kabra and Singh (2021) describe a method for designing peptides to inhibit the SARS-CoV-2 virus. They employed a ML algorithm (*Dead End Elimination*) to design a library of candidate peptides based on the diverse sequences, focusing on the main protease (Mpro) of the virus, crucial for viral replication. Computational simulations were used to virtually screen the peptide library, identifying four peptides with strong binding activity to Mpro. The study observed increased interactions between the peptides and Mpro, suggesting stable binding. Specific negatively charged aminoacids in Mpro formed strong interactions with all peptides, indicating their potential importance for inhibition. Four peptides emerged as potential candidates based on binding energies, conformation, and interaction strength with Mpro during simulations. The study underscores the advantages of using ML for peptide design due to its speed, accuracy, and ability to consider diverse sequences. The designed peptides showed promise for broad-spectrum activity against various COVID-19 strains, though further experimental validation in a lab setting is necessary to confirm their effectiveness against the virus.
In the study by Wang et al. (2022), the authors developed a *Retro Drug Design* (RDD) workflow to define the desired properties of drug molecules, including biological activity and drug-likeness properties. After establishing these desired properties, they employed the OptATP molecular descriptor system to generate molecular representations. A support vector machine (SVM) model was then used to predict the target properties of these molecules based on their molecular representations. To explore the space of possible molecular structures and identify those with the desired properties, they used the Monte Carlo sampling algorithm. Finally, they decoded the molecular structures from the solutions found by Monte Carlo sampling using a deep learning model called *LV2SMI*. This integrated approach allowed the authors to efficiently design drug molecules with specific desired characteristics.

Other authors have introduced sequence-based strategies for analysing protein sequences. For instance, Chen et al. (2023) employed ML to analyse protein sequences directly, bypassing the need for structural data. This approach leverages the readily available nature of protein sequences, offering significant advantages in speed and efficiency. Central to this method is TransformerCPI2.0, a model trained on a vast dataset of protein-compound interactions. The researchers validated TransformerCPI2.0 through three key approaches. First, they demonstrated its superiority over traditional methods in VS. Second, they developed tools for interpreting the model's predictions, which provided valuable insights into binding mechanisms. Finally, they successfully applied TransformerCPI2.0 to identify new drug leads for challenging targets that lacked structural information. The sequence-to-drug approach presents several benefits since it eliminates the bottleneck of requiring protein structures, accelerates the drug discovery process, and enables exploration of previously difficult targets.

Finally, the **VS of chemical compounds and the predictive modelling through ML techniques, significantly accelerate the pace of DTI by leveraging computational power to prioritise the most promising candidates for experimental validation.** In addition to that, some approaches, like molecular docking, can also predict the preferred orientation of a small molecule when bound to a protein target to form a stable complex. Evidence shows that available software tools can estimate the binding affinity through molecular dynamics simulations and molecular docking (Qureshi et al. 2023).

Box 3. Considerations from the experts: AI for DTI

Highlights resulting from Experts' consultation

According to one of the experts contacted for this report, the current revolution in AI for protein-based therapeutics is significant, particularly with the advent of generative AI. This technology enables the rational design of de novo proteins with tailored functions, properties, and molecular targets for a wide range of applications. This innovation is being backed by substantial industry investment, as demonstrated by companies such as X**aira, Generate, Isomorphic Labs, Absci, Dreamfold**, and others.

According to one expert form Academia, important data sets are derived from large genome studies and deep mutational scanning approaches. These techniques, which involve mutating every residue in a virus or viral protein, help to assess the impact on virus transmissibility and other characteristics.

Members from National regulatory Agency assert that repositories and libraries do not always include reliable data and suggested that conducting an evaluation of data quality needs to be included in libraries and repositories as a mandatory requirement.

The table below provides an overview of the main AI-based tools identified for DTI, the AI technique utilised, and their implementation in the market (**Table *3***).

Table 3. List of AI tools identified for DTI and compound screening

AI-based tool	AI Techniques	Functioning and goals	AI-based tool author	Availability[a]
TargetDB	ML	Extract and synthesize target-relevant information from a large volume of complex, disparate multi-omics data	De Cesco et al. 2020 quoted in Vijayan et al. 2022	Public
COVIDVS	Deep Learning	Identify potential drug candidates for treating COVID-19	Wang et al. 2021	Private
Dead End Elimination	ML	Design a library of candidate peptides based on the diverse sequences, focusing on the main protease (Mpro) of the virus	Kabra and Singh 2021	Private
Retro Drug Design - LV2SMI	Deep Learning	Define the desired properties for drug molecules, including biological activity and drug-likeness (ADMET) properties	Wang et al. 2022	Private
TransformerCPI2.0	ML	Analyse protein sequences directly, bypassing the need for structures	Chen et al. 2023	Private

[a] *For this study the availability of the tool in the market was defined as: "Research use" for AI-based tools developed within the academia but not yet implemented or tested within a real-world setting, "In development" for AI-based tools currently under design or improvement. Overall, the methodology of these tools has been published in the literature because of their innovative approaches, "Open source" for AI-based tools for which the original source code is made freely available and can be redistributed and modified, and "On the market" for AI-based tools that have been already deployed in real-world settings and are currently being used.*

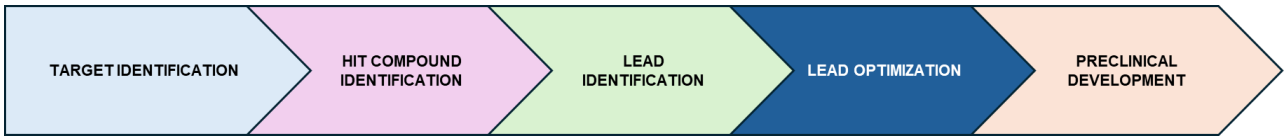
2.3. AI for pre-clinical research

The main outcomes of the literature review on AI-based tools for pre-clinical research are:

- AI is significantly enhancing the efficiency, speed, and success rates of pre-clinical drug development by streamlining target identification and validation. AI algorithms can process vast genetic and protein interaction data, prioritising promising targets and saving valuable time and resources.
- AI is crucial in assessing a drug candidate's potential efficacy and toxicity before proceeding to human trials. By analysing data from past clinical trials and pre-clinical research, AI can identify patterns predicting a drug's effectiveness and potential side effects, thus supporting researchers focus on candidates with the highest likelihood of success and the lowest risk of adverse effects.
- AI is instrumental in the development of MCM, including surveillance and diagnostics. By leveraging AI for the rapid detection and analysis of pathogens, researchers can improve outbreak response times and diagnostic accuracy, ultimately enhancing public health preparedness and response strategies.

A critical part of pre-clinical research is conducting rigorous safety tests to ensure the lead candidates are not toxic or unsafe before moving on to clinical trials in humans (phase III). Following the successful completion of pre-clinical tests, and if the results support the researchers' hypotheses, the clinical (in-human) studies can start to gain market approval and official launch (phase IV) (Sundari et al. 2024).
Traditionally, this phase has been a laborious and expensive process, full of uncertainties and limited by the volume of data that needs to be analysed. However, AI is revolutionising how we approach pre-clinical drug development, increasing efficiency, speed, and success rates, as previously mentioned**.** **Figure *2*** shows the different phases of the pre-clinical development workflow.

Figure 2. Pre-clinical development workflow



Source: Adapted from Vijayan et al. (2022)

One of the most significant contributions of AI in pre-clinical development lies in its ability to streamline target identification and validation, which was the focus of section 2.2. AI algorithms can process through vast repositories of genetic information, protein interactions, and disease aetiology to pinpoint potential target molecules - the specific molecules within a cell that play a key role in the disease process. This eliminates the traditional, time-consuming approach, allowing researchers to prioritise the most promising targets with greater accuracy.

Furthermore, **AI can analyse these potential targets, predicting potential side effects and the likelihood of their effectiveness against the disease** (Gadade et al. 2024). This crucial step helps to avoid dead ends early in the development process, saving valuable time and resources. As highlighted by Vijayan et al. (2022), through AI techniques, pharmaceutical companies can gain a deeper understanding of target biology, uncovering disease-target associations and identifying targets with a strong link to a specific disease. This leads to prioritising drug targets based on their potential for successful drug development, as in the case of TargetDB, a system that integrates publicly available data on a given target and uses an ML-based classification algorithm to categorise target tractability, providing useful criteria for protein ligandability assessment and prioritisation of drug targets for development.

Once a target molecule has been identified, the next step involves finding chemical compounds that can interact with the target and disrupt the disease's natural history. Evidence suggests that **AI-powered high-throughput screening allows researchers to analyse massive libraries of chemicals, predicting their potential to bind to the target molecule** (Sundari et al. 2024). This significantly reduces the time, and resources required compared to traditional screening methods that involve testing millions of compounds one by one and supports the subsequent phase, the lead optimisation phase, the most expensive and time-consuming phase in preclinical drug discovery (Vijayan et al. 2022). It involves iterative rounds of the design-make-test-analyse cycle and reducing the number of iterations is crucial for accelerating this process. In addition, generative model platforms that rely on AI-guided generative modelling for compound design have demonstrated success in reducing the number of iterations and designing compounds that meet the defined LO criteria (Vijayan et al. 2022). Current approaches rely on generative AI, the AI method used in large multi-modal models, to develop compounds with specific properties, as well as AI-based *de novo* protein therapeutics (WHO 2024).

Evidence shows that researchers are also training ML models to use in pre-clinical studies to predict the properties of a given molecule, such as its binding affinity to a target protein or its toxicity (WHO 2024). Before a drug candidate can proceed to human trials, it is essential to assess its potential for efficacy (its ability to produce the desired therapeutic effect under controlled conditions), effectiveness (how well it works in real-world conditions) and toxicity (harmful side effects). By analysing data from past clinical trials and pre-clinical research (including animal models and lab experiments), AI, especially ML and deep-learning subsets, can identify patterns that predict a drug's effectiveness and potential side effects. One specific that allows this kind of activity is the BenevolentAI's platform, already introduced in this report, which uses text mining techniques to identify potential treatments by analysing extensive datasets of gene expression, protein interactions, and clinical trial data to predict the efficacy and safety of drug candidates (Richardson et al. 2020). This allows researchers to make informed decisions about which drug candidates to pursue for further development, focusing their efforts on those with the **highest likelihood of success and the lowest risk of adverse effects** (Sundari et al. 2024). Predicting drug toxicity is crucial in drug development to ensure safety and efficacy, as mandated by regulatory agencies like the FDA and EMA.

Reviews on AI in toxicity prediction have highlighted its broad applications, diverse toxicity properties, and the need for comprehensive studies to optimise these models. The review of Van Tran et al. (2023) presents an overview of recent advances in AI-based drug toxicity prediction, including the use of various ML algorithms and deep learning architectures. Various ML methods, such as Random Forests (RF), Decision Trees (DT), k-nearest Neighbors (k-NN), and Support Vector Machines (SVM), have shown promise in accurately identifying toxic compounds in drug toxicity prediction. Deep Learning with the use of neural networks simulates the human brain's functioning, using multiple layers to extract features from input data and make predictions. Unlike traditional ML algorithms with simpler structures, deep learning models, based on complex artificial neural networks (ANN), require larger datasets and more computing power. deep learning models like Deep Neural Networks (DNN), ANN, Recurrent Neural Networks (RNN), Convolutional Neural Networks (CNN), GNN, Graph Convolutional Neural Networks (GCNN), and Graph Attention Networks (GAT) have been effectively used to predict drug toxicity, according to these authors, analysing molecular structures, identifying chemical properties, and leveraging patterns within biological datasets. These models process vast amounts of data from sources such as molecular fingerprints, gene expression profiles, and protein interactions. By learning intricate relationships and non-linear dependencies, they enable accurate predictions of adverse drug reactions, toxicity levels, and drug efficacy.

Efficient toxicity predictions have the potential to replace traditional pre-clinical approaches, such as in vitro and animal models. Moreover, these AI models can serve as risk management and prioritisation

tools in development pipelines by providing early indications of high-risk compounds with significant safety concerns (Askin 2023).

Translating pre-clinical discoveries into clinical practice in the form of new therapeutics is one of the biggest challenges as clinical candidates are lost during translation (Vijayan et al. 2022). The power of AI techniques to learn hidden and meaningful patterns by integrating large amounts of omics data sets provides valuable insights for translational biomarker discovery and is becoming pivotal. As innovations in AI technologies continue, the use of AI in pre-clinical research for drug development is expected to continue growing according to the available evidence (Vijayan et al. 2022). Finally, the study of Jayatunga et al (2024) underlines how the number of AI-discovered molecules in clinical trials has grown exponentially since 2015 with an 80–90% success rate in Phase I, followed by 40% in Phase II, substantially higher than historic industry averages, suggesting that AI is highly capable of designing or identifying molecules with drug-like properties. Initially, AI was mainly used to repurpose existing drugs, however, the trend is shifting towards discovering entirely new small molecules, with some success found in antibodies and vaccines. Besides drugs, AI has also been used in procedures to assess the characteristics of other MCM to ensure their quality and consistency during manufacture such as diagnostic tools. This includes including Chemistry, Manufacturing, and Controls (CMC) development, process control and development, process design and scale, autonomous systems for drug manufacturing, producing planning, management of assets, quality control, quality assurance, modeling for stability and shelf-life, management of documents, training, supply chain optimization, sustainability, labor and end-to-end operations visibility, among others (WHO 2024).

A recent FDA discussion paper (Food and Drug Administration, 2023b) debates how the incorporation of AI in advanced manufacturing techniques into pharmaceutical regulations has sparked extensive feedback from stakeholders, highlighting the transformative potential and challenges of AI in this sector. Firstly, stakeholders have identified a broad range of AI applications within the pharmaceutical manufacturing process. These applications include Chemistry, Manufacturing, and Controls (CMC) development, process control and development, quality assurance, and supply chain optimisation. Such diverse applications underscore AI's versatility and its potential to enhance various aspects of pharmaceutical manufacturing (Craven 2023). Also, the National Institute for Innovation in Manufacturing Biopharmaceuticals highlights the importance of understanding how AI can be used differently in manufacturing. It highlights that using AI to determine the ideal conditions for a process (defining operational parameters) is different from using AI to control those conditions during manufacturing. This distinction is crucial for understanding how AI can be effectively integrated into manufacturing processes (Craven 2023).

In summary, evidence found in the literature review suggest that research on AI used in supporting pre-clinical research is focused more on reviewing the use of various AI techniques and approaches rather than specific tools related to the pre-clinical phases.

Box 4. Considerations from the experts: AI for pre-clinical research

Highlights resulting from Experts' consultation

Members from Academia mentioned that since AI tools depend heavily on the quality of the data they utilise; the priority should be given to initiatives that focus on gathering and providing **high-quality data**. Equally important is the need for effective data storage and standardisation practices to ensure data is **accessible, searchable, and usable**. A significant amount of time and resources need to be dedicated to standardising data, locating relevant datasets, and ensuring their availability. The challenge often lies not just in building robust models, but in the fundamental task of data collection and organisation.

In drug discovery, public databases such as **ChEMBL** and **PubChem** are commonly used resources. However, the most valuable data often originates from pharmaceutical companies and remains confidential, limiting access for academic researchers and creating a gap in the available data landscape. While public databases are expanding, and promising initiatives like and some not yet published promising initiatives currently active at MIT to make chemical reaction data accessible are underway, much work remains to bridge this gap and enhance data accessibility for the broader research community.

According to experts from academia, **Generative AI** is revolutionising the rational design of drugs by targeting known structures or even generating structures of previously unknown targets, which can then be used for drug development. This approach is rapidly becoming a cornerstone in the field of protein-based therapeutics.

Foundational models offer the ability to design for species cross-reactivity, which allows testing across different species, such as mice and humans. This capability is critical when targeting host-specific molecules and is now routinely integrated into drug design to enhance the effectiveness of preclinical models, offering insight into both efficacy and toxicity. Additionally, these models are being used to design pharmacokinetic and pharmacodynamic (PKPD) parameters, such as the duration of drug action and bio-distribution. Efforts are also underway to design drugs with reduced immunogenicity, aiming to create therapies that trigger fewer immune responses. However, certain toxicity tests, such as those related to liver and kidney functions, will continue to require testing in animals and humans to ensure safety. Moreover, researchers are now currently experimenting the use of LLM-based tools for medical study reports, especially for the pre-clinical study, which historically take a long time to draft. With these tools, researchers can get a draft that's 80% complete within a few clicks. Ensuring that data is provided in the correct format is crucial, as it can significantly impact the accuracy and efficiency of data processing.

According to members of the industry, significant advancements in AI-driven drug development are anticipated in the coming years and the focus will be on enhancing virtual screening techniques and optimising molecular structures and molecules discovery. While the ability of AI to model complex biological systems is still developing, the adoption of more sophisticated ML algorithms is expected to drive rapid improvements. In addition, the increased speed and efficiency of AI in processing molecular data and generating actionable insights are likely to substantially reduce both the time and cost associated with drug discovery.

2.4. AI for clinical research

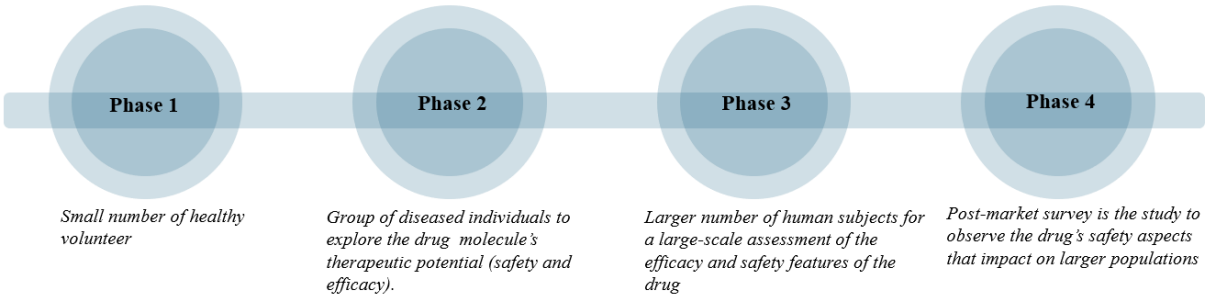
The main outcomes of the literature review on AI-based tools for clinical research are:

- AI enhances various aspects of clinical trials, including patient recruitment, therapy dosing, risk assessment and outcomes reporting. By improving patient matching, optimising treatment protocols, and predicting patient outcomes, AI significantly increases the efficiency and effectiveness of clinical trials.
- AI has the potential to both mitigate and exacerbate biases in clinical trials. Investments are necessary to address the digital divide and ensure privacy safeguards, which can help improve the diversity of trial cohorts and overcome barriers to equitable participation.
- Leveraging diverse data sources and advanced analytics though AI technologies is essential for advancing in personalised medicine and improving patient outcomes.

Once drugs or vaccines have passed the pre-clinical phases and are about to be introduced into a clinical trial, AI can be used to support or enhance various aspects of clinical development, design, recruitment, conduct and analysis. Phase I trials examine the drug candidate's pharmacokinetics and

safety in a small group of healthy volunteers, whereas phase II and III trials examine the drug candidate's efficacy and safety in larger patient populations (Sundari et al. 2024, Fultinaviciute 2022) (**Figure 3**).

Figure 3. The four main phases of clinical trials



Source: Adapted from Sundari et al. 2024

According to the literature, during the clinical trial phase, AI can be effective in different activities:

- Clinical trial design: analysing existing data to identify patterns and develop the most appropriate clinical design protocol
- Patient recruitment: selecting the right patient population;
- Therapy dosing: determining appropriate dosages and criteria for personalised treatments and real-time monitoring of the patient's response to treatments;
- Risk assessment and safety: optimising protocols with predicting modelling, identification of potential safety concerns, drug interactions, and early signs of efficacy
- Reporting: automating the generation of analysis reports and intelligently interpreting data for use in subsequent systems.

AI can significantly enhance clinical trial protocol development and management by boosting efficiency, accuracy, and cost-effectiveness. For instance, AI leverages data-driven insights by analysing large datasets from previous trials, electronic health records, and scientific literature to optimise trial design. This allows AI to identify and recommend the best inclusion/exclusion criteria, dosing schedules, and endpoints, ultimately improving the trial's success rate. Additionally, **AI-powered simulation models can predict patient outcomes across various trial designs, helping researchers refine protocols before actual trials begin**. Furthermore, AI-driven predictive modelling can anticipate potential challenges such as recruitment or retention issues, allowing for early adjustments to the trial design, and ensuring a smoother execution process.

Predicting patient outcomes in clinical trials is becoming increasingly sophisticated with the use of AI and ML. These technologies can improve precision medicine by reducing statistical variability and optimising clinical trial design improving its consistency (Mikkili et al. 2021, Askin et al. 2023). For instance, AI can support the creation of efficient statistical measures and identify participants who may progress quickly, potentially shortening trial durations. **AI can also analyse electronic medical records to predict which participants might drop out, allowing for targeted interventions that encourage longer participation and reduce the required sample size.** (Sundari et al. 2024).

In addition, AI can determine appropriate dosages and criteria for personalised treatments and real-time monitoring of the patient's response to treatments, improving treatment protocols in a much more efficient manner (Lv et al. 2024). For example, ML-based solutions have been found to consistently enhance and refine clinical trial design over time given that models are trained on a readily updated and

expertly curated dataset. Evidence suggests that it is important to take this into account as the efficiency and accuracy of ML-based pattern recognition and prediction can only be maintained if the training dataset is free of bias and a wide range of scenarios is well represented (Sreepadmanabh et al. 2020).

Regarding the analysis clinical data, AI supports the analysis of clinical data results and is able to integrate genetic, environmental, and lifestyle factors that can facilitate individualised treatment strategies. During the COVID-19 pandemic, AI-assisted telemedicine set a precedent for remote healthcare delivery, and AI applications and wearable devices enabled continuous monitoring of vital signs. The integration of lab data with patient history analysis proved invaluable during the pandemic, aiding vaccine development and understanding post-COVID-19 syndromes (Iqbal et al. 2023).

The research of Haas et al. (2021) explored the use of **Risklick AI**, an advanced data collection and analysis tool, in evaluating risks associated with various COVID-19 vaccine technologies. By aggregating data from multiple international sources, this tool assesses risks across categories such as safety, cost, manufacturing scalability, and shipping. An initial analysis in June 2020 identified unique strengths and vulnerabilities among DNA, RNA, inactivated viruses, virus-like particles, live attenuated viruses, protein subunits, and viral vector vaccines. A follow-up analysis in May 2021 revealed that few high-risk vaccines reached market authorisation, while RNA and protein subunit vaccines showed promising scalability and efficacy despite logistical challenges. This study underscores AI's crucial role in optimising vaccine development and highlights its potential to enhance future therapy innovations by providing detailed risk analyses and anticipating production and distribution issues that can accelerate the clinical trials phase in an unprecedented way.

Another study is the study of Abdulla et al. (2020) focused on **Project IDentif.AI**, an AI-based platform designed to optimise combination therapy for COVID-19 by rapidly identifying the most effective drug combinations and dosages. The study highlights the limitations of traditional trial-and-error methods in drug repurposing and dosing and showcases how AI can overcome these challenges by quickly identifying the most efficacious combinations. Within three days, Project IDentif.AI identified multiple highly effective and tolerable drug regimens, demonstrating the importance of precise dosing for optimal treatment outcomes. The study mentions the potential of AI in enhancing therapeutic development and provides insights into healthcare policies that could facilitate the broader deployment of such technologies to address COVID-19 and future pandemics.
Goldberg et al. (2024) highlighted the potential of decentralised clinical trials to enable patient participation from the convenience of their homes. By analysing extensive electronic health record datasets, AI can swiftly identify potential trial participants who meet specific criteria, accelerating recruitment and reducing associated costs. Furthermore, in this scenario, AI can predict which patients are most likely to benefit from a particular treatment, optimising patient selection and trial efficiency.

Overall, EMA outlines the expectations for using AI and ML in clinical trials within the framework of Good Clinical Practice (GCP), as specified in different guidelines (EMA 2023, 2023b). **These guidelines emphasise that all aspects of a clinical trial must adhere to strict standards to protect the rights, safety, and well-being of trial participants while ensuring that the data generated is credible and accurate**. Any AI/ML model developed for clinical trial purposes must be meticulously documented, which includes providing the full architecture of the model, detailed logs from the modelling, validation, and testing processes, and the training data used (EMA 2023). Additionally, a thorough description of the data processing pipeline must be included. These elements are key parts of the clinical trial data or the trial protocol dossier and must be available for comprehensive assessment during market authorisation or clinical trial application (EMA 2023). The EMA seeks to ensure that all aspects related to AI and health are meticulously documented, following the requirements of the **AI Act** (COM[2021] 206 final), which is now in effect and mandates the responsible implementation of innovative technologies.

2.4.1. Patient recruitment: selecting the right patient population

As reported by the WHO (2024), the integration of AI in recruiting for clinical trials presents both risks and opportunities regarding inclusivity. AI can increase the diversity of trial participants by using patient-matching algorithms to reach a broader cohort. However, achieving this goal will require additional investments, such as addressing the digital divide (the unequal distribution of access to, use of, or impact of information and communication technologies among different groups) and implementing appropriate privacy safeguards. **These investments can help overcome barriers to equitable trial participation, such as racial and ethnic disparities, lack of access to care, and financial constraints** (Fultinaviciute 2022). AI enhances recruitment and adherence in clinical trials by targeting specific patient groups with unmet needs which empowers pharmaceutical companies and scientists to make informed decisions throughout drug development and commercialisation, focusing on individuals likely to yield the best clinical outcomes. AI can also improve patient adherence and identify those who may benefit from a product, offering potential public health benefits (WHO 2024).

For example, a 2014 study revealed that 86% of clinical trial participants were white, and a 2019 study found that 79% of genomic data came from people of European descent. Bias regarding sex and gender is still prevalent in research and development, notably in the exclusion of minor and vulnerable population groups. **Enhancing diversity in clinical trials generally requires various measures, including legal mandates, incentives, and partnerships with non-profit organizations and other entities**. Recent progress has been made; for instance, in 2022, the USA enacted legislation requiring “diversity action plans” for clinical trials used by the FDA to determine a medicine's safety and efficacy (WHO 2024). Additionally, the WHO MS approved a resolution in 2022 to strengthen clinical trials, requesting the Director-General to provide guidance on best practices for non-state actors in designing and conducting clinical trials. This guidance aims to strengthen the global clinical trial ecosystem to meet the needs of major population groups, particularly focusing on under-represented populations (WHO 2024).

Algorithms can significantly enhance patient care by personalising interventions to improve retention and adherence, and adherence and determining optimal drug dosages for individual patients. For example, Roggeveen et al. (2022) conducted a two-centre randomised clinical trial involving critically ill patients with sepsis or septic shock. These patients were assigned to receive **AutoKinetics dosing**, a system that integrates population pharmacokinetic models with electronic health record (EHR) data.

Iqbal et al. (2023) elaborate on the concept of personalised medicine highlighting that it represents a method that considers an individual’s unique genetic makeup, environment, and lifestyle for treating and preventing diseases. It shifts away from traditional diagnostic methods that rely on shared patient characteristics, aiming to transform healthcare through prevention, personalisation, and precision. To achieve this, various data sources such as genomics, biological data, transcriptomics, and proteomics play a crucial role. Big data becomes essential in this process, and researchers are increasingly exploring the potential of AI for robust data analysis. This perspective underscores how leveraging diverse data sources and advanced analytics, including AI, is critical for advancing personalised medicine and improving patient outcomes. The authors (Iqbal et al. 2023) also mention AI tools that are making the difference in this sense, such as **DeepBind**, a fully autonomous standalone software designed to predict the sequence specificities of DNA and RNA binding proteins, or **DeepSea** (a deep learning -based sequence analyser), which employs deep learning techniques to forecast the chromatin effects of sequence alterations at the single-nucleotide level.

Box 5. Considerations from the experts: AI for clinical research

Highlights resulting from Experts’ consultation

According to experts from the pharmaceutical industry, in the field of pharmaceutical development, AI technologies, particularly NLP and LLM, are playing an increasingly important role. They assist in

patient selection, recording, and analysis of trial results based on study designs. Moreover, these tools are used to generate reports, analyse text, and support documentation for both development and clinical research. **Experts also mentioned that the ability of neural networks and ML algorithms to process large amounts of data is particularly advantageous for analysing existing pharmaceutical products with many batches**. However, potential biases in training sets remain a concern, especially given the strict GMP requirements. The initial deployment of AI appears most promising in automated defect analysis, particularly in sterile production environments where human operators are limited. While AI and ML are mainly used in pharmaceutical manufacturing, they also offer potential benefits for clinical trials. AI can optimise trial design, improve patient recruitment, and identify trends in trial data. However, challenges like data bias, transparency, and regulatory compliance must be addressed to ensure reliable and interpretable outcomes in both manufacturing and clinical settings.

On the other hand, according to the experts imaging analysis using neural networks has shown great potential in areas such as tumour detection and ocular analytics diagnostics, offering faster results than human analysis, though improvements in robustness are still needed as well as its implementation in other clinical field such as pathogen analysis which would be useful for health emergency management and crisis preparedness. AI-driven imaging analysis, such as convolutional neural networks or the algoriths from **Zebra Medical Vision** or the **CheXNet** by Stanford University, have also shown to be effective in detecting pneumonia from imaging. Additionally, AI can help identify biomarkers of systemic inflammation and sepsis, enabling early detection and intervention in critical patients. Some examples are the **Sepsis Detection Systems** (Duke University) to identify early signs of sepsis in ICU patients using ML or the **CureMetrix**, an AI to analyse health data and medical imaging data to predict biomarkers related to systemic inflammation. These advancements are especially valuable in health emergencies, where rapid diagnosis and timely treatment are essential for improving patient outcomes and managing resources efficiently.

Experts from European national regulatory agencies expressed their desire to enhance existing tools for identifying clinical trials that utilise AI, aiming to better match specific patients with the most suitable trials—something their current systems are not yet achieving. They also believe that AI could be leveraged to notify about side effects and answer questions related to specific products. Since much of the information these agencies handle is primarily text-based, they see NLP as the most appropriate AI technique to address these challenges.

As members of national agencies state, it is crucial to assess the safety of LLM and ensure that they are free from biases. A rigorous validation process with appropriate metrics is necessary to determine whether these AI systems can be implemented securely. For LLMs, efforts must be made to achieve semantic interoperability at the national, European, and international levels through data harmonisation and standardisation. Furthermore, the integration of LLMs and NLP into datasets (such as EUDAMET) could significantly enhance the identification and communication of side effects. National agencies must also be aware of the various AI tools available for use during health emergencies. The COVID-19 pandemic highlighted the need for enhanced training and awareness within administrative services, particularly in relation to the effective deployment of AI technologies in both clinical research and real-time healthcare decision-making.

Experts also mentioned that the use of AI tools for predicting drug efficacy and safety, particularly in toxicity prediction models, represents a significant promising but challenging area of research. Given that approximately half of clinical trials fail due to toxicity issuest, advancements in this domain could have a profound impact on drug development. **By improving the accuracy and reliability of toxicity prediction models, it would be possible to identify potential safety concerns at an earlier stage, well before reaching clinical trials.** This early prediction could significantly reduce the resources, time, and effort currently expended on drug candidates that are ultimately

unsuccessful due to toxicity. Therefore, there is a substantial potential for progress in the field of toxicity modelling, which could lead to more efficient and cost-effective drug development processes. As one expert from Academia noted, several organizations are currently working to advance our understanding of medicinal safety, such as the Cambridge Alliance on Medicinal Safety, as well as consortia aimed at fostering collaboration between academia and industry.

The table below provides an overview of the main AI-based tools identified to support the clinical research phase of MCM development, the AI technique utilised, and their implementation in the market (**Table 4**).

Table 4. List of AI tools identified for the clinical research phase.

AI-based tool	AI Techniques	Functioning and goals	AI-based tool author	Availability[a]
Risklick AI	ML	Evaluating risks associated with various COVID-19 vaccine technologies	Haas et al. 2021	Private
Project IDentif.AI	ML	Optimise combination therapy for COVID-19 by rapidly identifying the most effective drug combinations and dosages	Abdulla et al. 2020	Private
AutoKinetics	ML	Calculate a dose advice for any predefined dosing target	Roggeveen et al. 2020	Public
DeepBind	Deep Learning	Predict the sequence specificities of DNA and RNA binding proteins	Iqbal et al. 2023	Public
DeepSea	Deep Learning	Forecast the chromatin effects of sequence alterations at the single-nucleotide level	Iqbal et al. 2023	Public
Zebra Medical Vision algorithm	Convolutional neural network	Identify diseases and localise them within an image.	Zebra Medical Vision	Private
CheXNet	Convolutional neural network	Detect a variety of conditions, including pneumonia, from chest X-rays.	Stanford University	Open source
Sepsis Detection Systems	Convolutional neural network	Predict the onset of sepsis and monitor its progression in patients	Duke University	Private
CureMetrix	Deep Learning	Predict sepsis risk, as well as other inflammatory conditions, by identifying patterns in patient data that might go unnoticed by clinicians.	Retson et al. 2022	Private

[a]*For this study the availability of the tool in the market was defined as: "Research use" for AI-based tools developed within the academia but not yet implemented or tested within a real-world setting, "In development" for AI-based tools currently under design or improvement. Overall, the methodology of these tools has been published in the literature because of their innovative approaches, "Open source" for AI-based tools for which the original source code is made freely available and can be redistributed and modified, and "On the market" for AI-based tools that have been already deployed in real-world settings and are currently being used.*

2.5. AI for vaccine development

The main outcomes of the literature review on AI-based tools for vaccine development are:

- Pre-existing vaccines for SARS and MERS enabled a swift pivot to SARS-CoV-2, highlighting the importance of adaptable technologies for future pandemics.
- Recent ML advancements, including *in silico* techniques and neural network have shown promise in predicting molecular properties potentially aiding in the development of new vaccines.
- ML algorithms identify specific immune response targets, predict vaccine efficacy, and model antigen-antibody interactions, leading to more efficient and targeted vaccine development.

AI can be instrumental in vaccine development by analysing extensive datasets, identifying potential vaccine targets, and predicting the efficacy of vaccine candidates. According to the Coalition for Epidemic Preparedness Innovations (CEPI), **the advantages of AI-based approaches for vaccine development include accelerated development timelines, enhanced accuracy, and scalability.** Additionally, AI can contribute to cost reduction and improved vaccine safety by identifying potential adverse effects before clinical trials (CEPI, 2023).

The rapid development of COVID-19 vaccines underscored the effectiveness of AI tools in modelling, designing, and simulating the intricate interactions between drugs and human bodies, culminating in the creation of essential vaccines (Sreepadmanabh et al. 2020). Consequently, within just six months of the emergence of the previously unknown virus (SARS-CoV-2), early prototypes of potential vaccines were already undergoing testing (Gomèz-Gonzàles and Gonzàles 2023). The rapid development of COVID-19 vaccines has been attributed to the prior research on vaccines for SARS and MERS. Experts had been working on these vaccines before SARS-CoV-2 emerged, allowing them to swiftly adapt their knowledge and techniques to the new virus (Sharma et al. 2021). This approach not only accelerated vaccine development during the pandemic but also offered a valuable strategy for future outbreaks (Halabi et al. 2023). Additionally, experts emphasise that key scientific and technological innovations are crucial for developing effective vaccines quickly. These include enhanced surveillance and alert systems for early detection of viral outbreaks, as well as geographically diverse manufacturing capabilities to ensure equitable access to vaccines (Arora et al. 2021).

Historically, the development of new vaccines has been hindered by various factors, such as the dereplication problem in natural product discovery, where identical molecules are rediscovered multiple times, or the intrinsic limitations in chemical libraries that fail to accurately reflect the chemistry of antibiotic molecules, as mentioned by Stokes et al. (2020). **However, recent advancements in ML have shown promising advances in predicting molecular properties to identify new classes of antibiotics, which could be extended to vaccines**. *In silico* techniques, which use computer modelling or simulation rather than traditional experimental methods, enable the exploration of vast chemical spaces beyond current experimental reach. Evidence has shown that neural network-based molecular representations can automatically learn and map molecular fingerprints and descriptors, predicting their properties with greater accuracy than manual methods (Stokes et al., 2020).

Bravi (2024) showed that ML is emerging as an indispensable tool in the complex landscape of vaccine development. By examining vast datasets of immune responses, ML algorithms are unlocking a new era of precision medicine. A key focus is the identification of B and T cell epitopes - the specific molecular targets that trigger immune responses. This knowledge appears pivotal for creating vaccines that are not only effective but also highly targeted, potentially reducing adverse effects. Another innovative methodology was proposed by Chien et al. (2022) by using *in-silico* methods to design a multi-epitope vaccine. This method involves targeting specific viral proteins, which could lead to a more robust and comprehensive immune response. By predicting B and T-cell epitopes, researchers can create vaccines that address multiple aspects of the virus's structure and function (Chien et al. 2022).

The review of Sreepadmanabh et al. (2020) underlined the contribution of AI to the several approaches for vaccines against COVID-19. One promising strategy involved identifying pentapeptide sequences

unique to SARS-CoV-2. These sequences are expected to offer high immunogenicity and specificity, making them ideal targets for vaccine development. This approach, proposed by Guglielmo Lucchese (2020), highlighted a novel way to enhance the immune response against the virus (Lucchese 2020). Recent studies have employed immunoinformatics to pinpoint suitable target epitopes. For example, Baruah and Bose (2020) screened the viral surface glycoprotein for epitopes that can stimulate immune cells (Baruah and Bose 2020). Similarly, Ahmed et al. (2022) used insights from studies on SARS-CoV to identify conserved epitopes in SARS-CoV-2 (Ahmed et al 2022). These efforts demonstrate how computational analysis can guide the selection of effective targets for vaccine development.

Moreover, ML is proving knowledge at predicting vaccine efficacy and longevity by analysing data on immune responses across diverse populations which supports researchers at gaining valuable insights into how vaccines might perform in the real world. This predictive capability is crucial for optimising vaccine design and deployment strategies. At the core of these advancements, the evidence mentions lies the ability to model the intricate interactions between antigens and antibodies. For instance, ML algorithms are learning models that can recognise patterns and produce information that can be used to prioritise potential vaccine candidates. According to the evidence, this data-driven approach is transforming the traditional trial-and-error process of vaccine development into a more efficient and targeted one. To achieve these breakthroughs, ML models are being trained on massive datasets encompassing immune repertoires and the molecular composition of pathogens (Bravi 2024). This wealth of information allows algorithms to learn complex patterns and make accurate predictions that produce insights which are not only accelerating vaccine development but also deepening our fundamental understanding of the immune system (Cihan and Ozger 2022).

Another example of how AI is supporting the research on immunity has been reported by Haas et al. (2021) where the authors emphasised the capability of AI to predict antigenic peptides presented by major histocompatibility complex molecules, which are pivotal for vaccine-induced immune responses. Network-based algorithms have become increasingly vital in vaccine development which have been used to identify potential epitopes using techniques such as deep neural networks, gradient-boosting decision trees and artificial neural networks. Programs such as **MARIA** and **MoDec** leverage ML to predict antigen presentation, thereby facilitating vaccine development (Haas et al. 2021).

AI-based tools have also analysed the presentation of SARS-CoV-2 viral peptides to comprehend natural immunity and aid in the design of effective vaccines. For instance, **Vaxign**, a ML-based tool, was employed to predict vaccine targets. Expression-based algorithms, such as **ProteinMPNN**, have also been crucial in vaccine development as they can optimise the immunogenicity and expression of vaccines by enhancing the sequence to improve efficacy and safety. Another example is the COVID-19 vaccine developed by **Icosavax**, which represented the first *de novo* protein therapeutic to the clinic, a vaccine fully designed using computational methods[2]. One commonly used expression-based technique in mRNA vaccine development is codon optimisation, which increases stability, safety, and efficacy. According to the evidence, this optimisation is achieved by adjusting the balance between host tRNA availability and codon usage frequency, thereby improving translation efficiency and in vivo expression of the target antigen (Ong et al. 2021). As reported by Lalmuanawma et al. (2020) researchers from Taiwan are building a new model to improve the development of novel vaccines for COVID-19, by applying the ML and AI technology-based model on two datasets (one using the 3C-like protease constraint and other data-holding records of infected SARS-CoV, SARS-Cov-2, influenza, and human immunodeficiency virus (HIV)) using deep neural network on the eighty old drugs with potential for Covid-19 treatment; the study suggested eight drugs, i.e., vismodegib, gemcitabine, clofazimine, celecoxib, brequinar, conivaptan, bedaquiline and tolcapone are found virtually effective against feline infectious peritonitis coronavirus.

[2] For some information visit: https://icosavax.com/pipeline/

Additionally, evidence also suggested that integrated docking simulation algorithms provide an effective method for predicting ligand-target protein interactions. These algorithms can achieve precise predictions of molecular interactions through VS and multiple docking tools. Integrated docking simulation algorithms like PatchDock, AlphaFold or RosettaFold have provided the dynamic characteristics of designed systems and complexes by simulating the molecular docking environment. PatchDock has been used to dissociate docked molecules to form protein-protein complexes and can accurately predict global properties such as hydrogen bond formation and molecular conformation in complex formation (Ghosh et al. 2023).

The main organisations focused on vaccine development, such as GAVI or CEPI, prioritise developing vaccine libraries. These libraries consist of the prototypes of vaccines against representative viruses from various families, which can be swiftly adapted to combat-related viruses. For example, according to CEPI, in the last century, we have faced three epidemics involving coronaviruses: SARS, MERS and COVID-19. Although caused by different viruses, these diseases belong to the same family and show similar behaviours (CEPI 2023). A practical application of this approach is the use of rapid response platforms like mRNA or the **ChAdOx platform**, used in the Oxford/AstraZeneca COVID-19 vaccine. By advancing these prototypes through pre-clinical and early-stage clinical testing, robust safety and immunogenicity data can be built. Libraries allow for an early response when a new virus emerges, enabling vaccines to be quickly adapted, as happens with the annual update to the flu vaccine (CEPI 2023). The WHO has convened groups of global experts to prioritise pathogens within each virus family, so their recommendations can focus the vaccine library investments, which can represent an important step forward.

AI tools have also been used to design effective vaccines and evaluate their safety and acceptance among the population. During the COVID-19 pandemic, AI also supported governments to assess public perception of COVID-19 vaccines and raised vaccine awareness. By analysing previous data, AI predicted disease progression and suggested vaccine development paradigms. For instance, databases like Corona-DB-AI compiled COVID-19-related compounds and peptides for training models in treatment extraction (Arshadi et al. 2020). MIT's computer science and AI lab developed AI programs OptiMax and EvalMax, which designed peptide vaccines and assessed their efficacy across racial and minority populations. Their study suggested that adding certain peptides to the Spike protein may enhance immune responses (Arora et al. 2021).

Box 6. Considerations from the experts: AI for vaccine development

Highlights resulting from Experts' consultation

Experts from various disciplines agree that the application of protein design for developing new vaccines has become highly significant, especially in areas like dengue fever. Deep learning approaches to protein folding and design are enabling the creation of novel vaccines in several ways:

1. **In silico protein folding:** Previously unstructured antigens can now be folded computationally, with the resulting models used for vaccine development.
2. **Redesign of antigens:** Antigens that were historically difficult to fuse or scaffold can be partially redesigned to improve their solubility, stability, and facilitate their fusion with common viral vaccine vectors.
3. **Epitope evaluation:** Antigens can be evaluated in silico to identify the best epitopes, maximising the potential immunogenicity of the vaccine.

Another researcher added that, thanks to AI tools like RoseTTAFold and AlphaFold, it is now possible to predict protein structures, enabling rapid in silico responses where researchers can immediately optimise antigens for vaccine development. AI-based solutions have eliminated the need to solve the glycoprotein structure of viruses causing an outbreak. Some research institutions in the US are now

designing small binder proteins or antibodies that can target the virus purely based on the predicted structure, bypassing the need for the actual structure. This approach can save valuable time, reducing the process from around six weeks—such as the time it took to solve the structure of the COVID-19 virus to approximately six hours for creating small binder designs.

Another promising area identified by academic experts is the design of macrocyclic peptides and antibodies, which can now be developed using computational tools. Experts highlighted that when validating AI models for diseases that are either neglected or have low prevalence, they often use analogous diseases to test biological products.

When discussing the validation of AI models for rare diseases, experts emphasised the importance of developing robust and reliable models. Senior researchers in the field of emergency response noted that during fast-moving outbreaks, vaccines are not always a viable solution, as protective immunity can take four to six weeks to develop. In such scenarios, biologics play a crucial role, though this field remains under-optimised.

One expert also provided examples of how biologics could be used in health emergencies: if biologics were available to provide immediate protection to someone exposed to an infectious disease, such as healthcare workers, or to conduct ring administration to contain an outbreak. She noted that AI has the potential to unlock significant advancements in this field, though a comprehensive toolkit for its development has yet to be established.

Members from national agencies reveal the COVID-19 pandemic, there has been a shift towards developing vaccines based on RNA technology. This approach underscores the importance of identifying surrogate biomarkers that could enable shorter and more streamlined clinical trials, facilitating the rapid development of potential vaccines. To achieve this, comprehensive data on current vaccines under development is essential, and AI could play a crucial role in analysing and leveraging this information. However, the effective implementation of such AI tools would require a well-defined regulatory framework to ensure their safe and reliable use in the vaccine development process.

The application of AI algorithms in the development of vaccines and drugs presents distinct challenges and considerations. While the utility of these algorithms can be enhanced, it is crucial to acknowledge the inherent trade-offs between complexity and interpretability. Experts in the field are continuously working to improve the quality, reliability, and performance of these models, aiming to strike a balance that maximises their accuracy and utility.

However, effectively utilising these algorithms requires a tailored approach, as a one-size-fits-all solution is not feasible. Many current algorithms lag behind in terms of development, facing specific challenges that must be addressed to improve their effectiveness and reliability. Addressing these challenges is essential for advancing the capabilities of AI in both vaccine and drug development.

The table below provides an overview of the main AI-based tools identified to support vaccine development, the AI technique utilised, and their implementation in the market (**Table *5***).

Table 5. List of AI tools identified for vaccine development.

AI-based tool	AI Techniques	Functioning and goals	AI-based tool author	Availability[a]
MARIA	ML	Predict antigen presentation, thereby facilitating vaccine development	Haas et al. 2021	Public

MoDec	ML	Predict antigen presentation, thereby facilitating vaccine development	Haas et al. 2021	Public
Vaxign	ML	Predict vaccines' target	Ong et al. 2021	Public
PatchDock	ML	Provide the dynamic characteristics of designed systems and complexes by simulating the molecular docking environment	Ghosh et al. 2021	Public
OptiMax, EvalMax	ML	Identify the relevant peptide and designing peptide vaccine	Arora et al. 2021	Public

[a] *For this study, the availability of the tool in the market was defined as: "Research use" for AI-based tools developed within the academia but not yet implemented or tested within a real-world setting and "In development" for AI-based tools currently under design or improvement. Overall, the methodology of these tools has been published in the literature because of their innovative approaches, "Open source" for AI-based tools for which the original source code is made freely available and can be redistributed and modified, and "On the market" for AI-based tools that have been already deployed in real-world settings and are currently being used.*

2.6. AI for automation and optimisation

AI holds a significant potential to support and transform the drug and vaccine development process through optimisation and automation across various stages of development:

In drug discovery and design, AI-driven molecular modelling and simulations, especially deep learning, can predict molecular structures and interactions, simulate protein-ligand interactions, and design new drug candidates that have a higher likelihood of success and support lead compound identification by screening massive compound databases using predictive models to identify leads that may bind effectively to specific biological targets.

During preclinical testing, AI offers predictive toxicity models to assess safety early, reducing reliance on animal testing. It also supports drug formulation by optimising the chemical properties of drugs, ensuring they are suitable for human use. Finally, in this phase, AI techniques contribute to the automation of lab processes, such as integrating robotic systems which can handle repetitive lab tasks like screening compounds and conducting high-throughput experiments.

Clinical trial optimisation benefits from AI in patient recruitment, personalised dosing, and real-time monitoring, leading to more precise treatments. For example, AI can analyse patient data to identify suitable candidates for clinical trials based on genetic, demographic, and medical history, ensuring faster and more precise recruitment. The analysis of such data can also lead to optimising dosages based on factors like genetics and metabolic rates, leading to more effective and personalised treatment plans and real-time monitoring and predictive analytics through AI-driven platforms can monitor patients in real-time, predicting potential adverse events or deviations from expected outcomes, allowing for rapid interventions.

In vaccine development, AI can assist in identifying and designing vaccine antigens by analysing viral genomes and predicting epitopes that elicit strong immune responses. ML models have also been demonstrated to be able to speed up vaccine development by identifying adjuvants and optimising their combination with the antigen. Particularly in health emergencies, AI can provide rapid vaccine development with AI tools that can analyse viral mutations and predict their effects in a short time, accelerating the design of vaccines to address new strains or emerging pathogens.

Furthermore, AI enhances clinical data analysis and post-market surveillance by including data-driven decision-making through the analysis of massive datasets from clinical trials, identifying patterns and outcomes faster than traditional methods, leading to quicker decision-making and monitoring real-world data from electronic health records, social media, or wearable devices to detect any long-term side effects of drugs or vaccines and provide insights into their efficacy and safety in diverse populations.

2.7. Limitations and challenges for AI in MCM development

When discussing the limitations and challenges of AI implementation in drug development, it is essential to address several critical issues, particularly around **data quality**, **bias**, **privacy**, and **transparency**. Drawing on insights from Vayena et al. (2018), as well as broader ethical and regulatory considerations, the following points are essential to understanding the challenges AI faces in this context.

The results of the consultation activities conducted for this Report revealed three major areas of limitation and challenges in the field of AI for medicines discovery and development.

- **Data quantity, quality and representativeness**: insufficient availability of high-quality data and lack of diversity, which can lead to biased or inaccurate outcomes when applied in AI models.
- **Data security, privacy and ethics**: sensitive medical data raise serious concerns around security and privacy, with potential vulnerabilities to breaches. Ethical issues, such as the proper use and consent of patient data, also pose significant challenges.
- **Regulatory challenges and the need for solid frameworks**: There is a pressing need for comprehensive guidelines to ensure that AI applications in medicine are implemented safely, ethically, and effectively.

A key element of AI infrastructure is input data, as the effectiveness of AI-driven advancements in drug discovery largely relies on the quality and quantity of data used to train and evaluate AI algorithms, along with several other factors. A major limitation of AI in DD is the “data-hungry nature” of AI models, particularly deep learning, which depends heavily on the quantity and quality of training data. Gangwal et al. (2024) suggest several techniques to overcome this **data scarcity**, such as transfer learning, active learning, one-shot learning, multi-task learning, data augmentation, and data synthesis, approaches that are under validation and in continuous evolution. Another significant challenge is data silos, where critical biomedical data is spread across organisations, limiting collaboration and indirectly affecting data availability; data silos can result from various factors such as proprietary interests, differing data formats, regulatory restrictions, and privacy concerns. As a result, valuable data is often inaccessible for broader use, preventing researchers and companies from gaining comprehensive insights (Gangwal e al. 2024). Linked to this, the experts consulted highlighted that an organisation's computational capacity is crucial for processing the large volumes of data involved.

Another significant limitation of AI in drug development is the **quality and representativeness of the data** used to train ML models. In drug discovery, the datasets must represent diverse populations to ensure the efficacy of drug candidates across different demographic groups. However, training datasets may suffer from biases, either because they overrepresent or underrepresent certain populations or because they reflect systemic biases in healthcare. For example, Vayena et al. (2018) highlight how medical datasets have historically shown biases, such as the overdiagnosis of schizophrenia among African Americans or the overrepresentation of older white males in certain medical studies. This raises concerns that AI algorithms may inadvertently perpetuate or exacerbate health disparities if these biases are not adequately addressed.

The need for large datasets, including those that span diverse patient populations and medical conditions, brings to the forefront **ethical concerns related to data privacy and consent**. AI models rely on vast amounts of personal health data, often requiring the integration of data from various sources. These include clinical trials, electronic health records (EHRs), and even consumer health apps (medical devices used to monitor people's health conditions). However, different jurisdictions have different regulatory frameworks, such as the **European General Data Protection Regulation (GDPR)** and the **Health Insurance Portability and Accountability Act (HIPAA)** in the U.S., which govern the use of health data. As Vayena et al. (2018) note, GDPR is more stringent and comprehensive than HIPAA, which does not cover health data generated outside of traditional healthcare settings (e.g., through wearables and health apps). On this point, it is necessary to underline that privacy is a growing concern for individuals who fear a loss of control over their personal health data in an increasingly data-driven healthcare and clinical research environment (Vayena et al. 2017). New regulatory models that include data portability rights (allowing individuals to manage data access), new mechanisms for informed consent (to control data use), and participatory governance schemes (involving individuals directly in data governance) can help with the perception that big data diminishes personal control (Vayena et al. 2017).

The implementation of **regulatory standards in AI development**, particularly within fields involving sensitive data such as healthcare, presents a multi-dimensional challenge. On one hand, developers are tasked with ensuring compliance with strict regulatory frameworks that govern how patient data can be collected, stored, and used. These regulations, such as GDPR or AI Act, are designed to protect individuals' privacy and ensure that data usage is ethical and consensual. However, they add a significant layer of complexity to AI development, especially as the legal landscape is continuously evolving and varies across jurisdictions. Maintaining the scalability and flexibility of AI models under these conditions becomes a difficult balancing activity. Scalability requires models to process and learn from vast amounts of data, but regulatory constraints often limit the types of data that can be used or require significant effort to anonymise or secure that data. In cases where patient consent is required for data usage, ensuring that such consent is valid and traceable across diverse systems becomes even more challenging. Furthermore, different countries or regions may have unique interpretations of what constitutes proper consent or data usage, meaning AI developers must create systems that are adaptable across multiple legal environments without compromising their operational scale(Vayena et al., 2018).

Beyond privacy concerns, AI developers must also handle the diversity of data formats and the specific conditions for their use. Medical data, for example, comes in a variety of formats ranging from structured data like lab results to unstructured data such as doctor's notes and imaging files. Each data type may have specific legal and ethical considerations attached, and ensuring that AI systems can process these data while remaining compliant adds to the implementation complexity. Additionally, data might be subject to restrictions based on its origin, the type of medical condition it pertains to, or even the demographic group it belongs to, further complicating the process of creating flexible AI systems that can scale effectively while staying within the bounds of legal requirements (Vayena et al., 2018).

To address these challenges, AI developers often need to integrate robust data governance frameworks that automate compliance processes, implement secure data-sharing practices such as federated learning to reduce the need for data centralisation and utilise advanced encryption and anonymisation techniques to safeguard patient privacy. However, these solutions must be carefully designed to avoid hindering the performance or scalability of AI models, underscoring the inherent tension between regulatory compliance and the operational goals of AI systems (Vayena et al., 2018).

As one policy expert assessed, economic and organisational challenges also play a significant role in the development and deployment of AI solutions in healthcare. The high costs associated with AI development can be prohibitive, especially for small and medium-sized enterprises (SMEs). This

situation underscores the need for increased venture capital investment, improved technology transfer offices, and supportive infrastructures that can foster innovation without stifling competition. Establishing supportive ecosystems is crucial for enabling SMEs to contribute meaningfully to AI advancements in healthcare, ensuring that innovation is not limited to large corporations alone. Furthermore, it is always important to update the policies regulating the use and development of technology in healthcare to provide technological innovation and industries with efficient guidelines, as the EMA has done in recent years (EMA 2023b, 2023c). As policy experts assess, there is a need for improved and increased dialogue between different stakeholders to facilitate the introduction of innovative products, including those incorporating AI, into the market and ultimately to patients. At a regulatory level, improved collaborations can help determine whether a product falls under the medical devices' framework or the medicinal products' legal framework, highlighting the need for a more streamlined and accessible point of contact for such guidance.

2.7.1. Regulations on AI for Drug development

In Europe, the use of AI technology in the field of health can be considered as a combination of different regulatory frameworks designed to safeguard patient privacy, ensure safety, and promote ethical standards. At the heart of this regulatory landscape is the **General Data Protection Regulation (GDPR)**, which governs how personal data is collected, stored, and used. Given the sensitive nature of health data, GDPR imposes strict requirements on healthcare AI systems, particularly around obtaining explicit and informed consent from patients. It also mandates principles like data minimisation, meaning only the necessary data should be used, and provides patients with important rights over their personal information, including the right to access, correct, and even request the deletion of their data. Compliance with these privacy standards is essential for any AI system operating in the healthcare sector (Amini et al., 2023).

Beyond data protection, the **European Health Data Space (EHDS)** is an emerging initiative that aims to facilitate the safe and secure sharing of health data across Europe. While still being developed, the EHDS has the potential to significantly advance the field of AI in healthcare by enabling access to vast amounts of high-quality health data, which is critical for training and improving AI models. At the same time, it ensures that data is shared ethically, with strong protections for patient privacy and consent, making it a crucial component of Europe's vision for AI-driven healthcare innovation (Horgan et al. 2022).

When it comes to AI systems that are classified as medical devices, the **Medical Device Regulation (MDR)** plays a pivotal role. MDR came into effect in 2021 and requires any AI tool used for diagnosis, monitoring, or treatment to meet stringent safety and performance standards. This includes demonstrating clinical efficacy and undergoing rigorous testing before being approved for use. Importantly, MDR also mandates continuous post-market surveillance, ensuring that these AI systems perform safely and effectively once deployed in real-world healthcare settings (Beckers et al. 2021).

In addition, the **EU Artificial Intelligence Act** represents Europe's effort to provide a comprehensive regulatory framework specifically tailored to AI technologies. This Act introduces a risk-based approach, classifying AI systems into various categories based on the potential risks they pose to individuals. Given the critical nature of healthcare, AI systems in this domain are often classified as "high-risk" and must adhere to strict transparency, accountability, and AI governance requirements. The AI Act also emphasises the need for ongoing monitoring and human oversight to ensure that AI systems remain reliable and safe over time.

Finally, the **Ethics Guidelines for Trustworthy AI**, developed by the European Commission's High-Level Expert Group on AI, provide an overarching ethical framework to mitigate the risks given by AI systems. Although non-binding, these guidelines emphasise the importance of human agency, transparency, and fairness in AI systems. In healthcare, this means that AI should always operate under

human oversight, be explainable to users and stakeholders, and avoid perpetuating biases that could lead to unequal treatment of patients.

In an interview with an expert from the Industry, several key points were raised about the integration of AI in the pharmaceutical industry, particularly regarding regulatory challenges, especially for what regards LLM and NLP tools. A significant challenge lies in navigating the complex regulatory environment in the EU, where regulations exist at both the national level and the EMA level. Pharmaceutical companies must not only meet EMA's regulatory requirements for AI tool usage but also ensure compliance with varying national regulations across the countries in which they operate. The introduction of the AI Act has added another layer of complexity, as it intersects with existing medical regulations, creating a need for greater clarity on how these frameworks align. Additionally, there is an ongoing concern about risk management, particularly regarding potential biases introduced by the quality of data used in AI models. To address these issues, Roche and the EMA engage in regular discussions, particularly through EMA's Quality Innovation Group, to evaluate the implications of AI in drug development practices and explore solutions for safe and effective integration of AI technologies.

Lastly, experts from European national regulatory agencies emphasised the importance of having a clear regulatory framework. They highlighted that once the AI Act is approved, it will be crucial to understand the specific requirements for AI-based software and tools. This is essential to ensure that the decisions made by these systems are robust and grounded in quality data, especially given their potential direct impact on patient treatment.

2.8. AI to support decision-making on MCM development

As stated by the WHO, AI could provide broader public and global health benefits. This requires the intentional use of AI to address unmet needs or to strengthen aspects of drug development or access that are neglected in the current pharmaceutical system. It also requires supportive investments, policies and practices through proactive action by the public sector, philanthropies and not-for-profit organisations (WHO 2024, p. 11).

At the European level, the EU is developing the European Health Data Space, a framework for health data sharing with clear rules, standards, and governance. This initiative aims to keep individual data within national borders in TRE-like infrastructures, using "data federation" technology to analyse multi-country data while preserving privacy (Hussein et al. 2024). Global standards for data-sharing and governance could enhance these efforts but require strong cooperation among various stakeholders.

Another concern relies on the use of AI to accelerate the development of medicines, which may increase a medicine's effective patent life (effective patent life is the period of patent protection remaining for a drug at the time of regulatory approval), and therefore the total time a company can charge high prices and maximise revenues. This could accentuate the tendency of companies to get to the market as quickly and inexpensively as possible by addressing the most profitable indications or population groups and leaving aside less represented indications or populations, such as children or populations with multiple diseases due to the requirement for additional clinical development after approval which could represent an overall greater public health benefit. Furthermore, speed might be prioritised over the quality of R&D, including the reproducibility of results. However, it has been observed that the quality of R&D decisions, including which compounds to take forward and the design of clinical trials, can reduce the failure rate and, therefore, have the most significant impact on the overall value of the project compared to reducing the cost or time of a particular phase (Druedhal et al. 2024). A practical example is how AI, in particular ML, has supported and transformed the management of seasonal influenza thanks to the analysis of patterns and disease chains to develop the most adapted vaccine, which could potentially be extended to other infectious diseases (Shah et al. 2024).

Box 7.*Considerations from the experts: AI to support decision-making on MCM development*

Highlights resulting from Experts' consultation

According to the consulted experts, AI plays a crucial role in supporting decision-making throughout the entire lifecycle of drug and vaccine development. Experts from governmental organizations and academia agreed that one of the fields were AI could support is in epidemiological surveillance of future outbreaks by predicting how diseases are spread and aiding public health organizations in identifying which vaccines and drugs will be needed. In this regard, AI can also support in predicting which viral strains are most likely to require vaccines, supporting researchers in being updated of rapidly evolving pathogens. An expert from Academia also mentioned that AI can support the identification of adjuvants, therefore, optimising vaccine efficacy.

In pre-clinical research, experts highlighted that AI techniques, particularly predicting models, can support the understanding of how diseases might progress in individuals and thus are crucial in developing personalised treatments. In addition, ML models have also been identified to predict side effects and toxicities before they reach human trials, potentially reducing the risk of failures at later stages and supporting the tasks this phase needs during drugs or vaccines development.

As mentioned by the available evidence, experts agree that AI is currently playing a crucial role in enhancing clinical trials through patient recruitment and trial design optimisation. Experts from a National regulatory Agency highlighted that, particularly in the recruitment phase, AI helps to identify suitable participants by analysing medical records, genetic data, and other relevant factors to match individuals with specific trial criteria. This ensures that the right candidates are selected, increasing the likelihood of successful outcomes. Additionally, experts emphasised AI's capacity to contribute to trial design optimisation by simulating various trial designs, predicting potential outcomes, optimising dosages, and reducing patient dropout rates. These advancements were noted to make clinical trials more efficient and effective, leading to faster and more accurate results.

3. Discussion and conclusions

In this study, we investigated the application of AI-based tools in the development of MCM, relevant to preventing, preparing and responding to health emergencies. The main goal of this report was to understand how AI technology can enhance MCM development, pre-clinical and clinical research, improve clinical trial design, and support decision-making throughout the MCM development process, in the context of health emergencies. By conducting a comprehensive literature review and engaging in consultation activities, we assessed the current state of AI in drug discovery and vaccine development, as well as how AI can support clinical trials, data analysis and identification of key targets during pre-clinical research. In addition, we have identified key challenges and explored their implications for health preparedness and response.

The rapid advancement of AI in healthcare presents both significant opportunities and complex challenges, especially in the development of MCMs. A critical aspect highlighted by the results is the importance of data quality and standardisation. AI models in healthcare require large volumes of high-quality, annotated, and interoperable data to function effectively, and without standardised data collection and curation processes across the industry, the reliability and consistency of AI applications can be compromised. **Thus, establishing robust data standards is essential to ensure that AI models produce accurate and trustworthy outcomes.**

AI's integration into the drug discovery pipeline represents a transformative shift in the pharmaceutical industry, driving significant advancements by predicting protein folding, molecular interactions, and cellular disease mechanisms. Advanced algorithms and data analytics enable the identification of novel drug targets, the design of promising compounds, and the optimisation of clinical trials. This technology accelerates the early phases of drug development and reduces risks by forecasting potential adverse effects. Additionally, AI enhances personalised medicine by analysing patient data to tailor treatments to individual genetic and health profiles, thereby maximising efficacy and minimising side effects. The ongoing evolution of AI-driven data management and automation further streamlines research workflows, cuts costs, and speeds up the delivery of new therapies. As AI technology advances, its role in drug development is expected to expand, offering new opportunities for innovation and precision in addressing complex diseases.

Our findings reveal that the fast-evolving domains of AI, particularly ML, demonstrate significant potential to enhance every stage of the medicinal product lifecycle. Current regulatory principles, guidelines, and best practices apply to AI, and organisations should integrate data science expertise with medicinal development and pharmacovigilance efforts in the coming years. Literature also indicates significant progress in drug discovery and repurposing, with AI techniques previously applied to drug repurposing now extending to other areas, such as small molecule discovery, vaccines, and antibodies. This evolving synergy between AI and drug development promises more effective, safer, and personalised treatments, paving the way for notable improvements in global health outcomes. The ability of AI, particularly ML, to accelerate development processes is crucial during health emergencies. By integrating AI with pharmacovigilance and regulatory frameworks, organizations can strenghten global health preparedness and achieve ensuring better outcomes during crises like pandemics.

Regulatory challenges are another significant concern in deploying AI within the healthcare sector. There is an urgent need for adaptive legislation and regulatory frameworks, such as regulatory sandboxes, to keep pace with the rapid evolution of AI technologies. Regulatory bodies like the EMA and the FDA must enhance their capacity to evaluate AI tools effectively. This includes not only understanding the AI technologies themselves but also recognising that continuous advancements in AI can lead to improvements in algorithms, software and tools over time. Experts agreed that regulations should account for this dyamic nature by incorporating flexibility during assessments throughout the

product lifecycle. Implementing adaptive regulatory measures is crucial for facilitating innovation while ensuring patient safety and maintaining public trust.

Ethical and security considerations are at the forefront of discussions about AI in healthcare. Ethical challenges include issues of bias and discrimination within AI models, which can arise from the data used to train these models or from the teams developing them. Contextual biases, depending on where and how AI recommendations are applied, also pose significant concerns. Data privacy and security threats, such as cyberattacks and the potential misuse of AI-generated information, for example, in spreading sensitive and personal data, highlight the need for advanced data encryption techniques and stringent security protocols. **Enhancing and evolving regulatory frameworks across MS to accommodate the emerging role of AI in the near future is essential to ensuring that decisions made using AI tools are both robust and free from adverse effects on citizens**.

Collaboration between academia, industry, and government is identified as both a necessity and a challenge. **Although open innovation plays a crucial role in advancing AI technologies, companies frequently encounter legal and competitive challenges that hinder collaboration**. Challenges such as determining the right moment to become exclusive and safeguarding trade secrets while sharing data pose significant obstacles, or striking a balance between promoting innovation through collaboration and preserving competitive advantages is crucial. Achieving this balance is essential for fostering an environment where AI can flourish without jeopardising individual or corporate interests.

In terms of fields of experimentation, several areas are experiencing current challenges and hold significant expectations for the near future. In medical devices and wearables, integrating AI faces limitations of interpretability and interoperability with existing healthcare workflows. However, advancements are expected in personalised healthcare solutions, necessitating that regulatory bodies adapt their evaluation processes for AI-integrated devices. The goal could be moving towards a smooth incorporation of AI into medical devices to an extent that enhances patient care without introducing new risks.

Another topic raised by this report is the use of AI-based tools to support the design and optimisation of clinical trials. **Leveraging AI and ML holds the promise of accelerating drug development while enhancing the safety and efficiency of clinical trials.** However, it is crucial to evaluate whether AI introduces specific risks and potential harms. As noted, AI algorithms can magnify errors and existing biases present in the data, which raises concerns about their generalisability and ethical implications when findings are applied beyond the initial testing environment. Additionally, the complexity of AI systems can lead to limited explainability or opacity due to proprietary reasons, posing challenges for transparency.

To address these issues, **there is a current focus on developing standards for trustworthy AI that emphasises characteristics such as explainability, reliability, privacy, safety and security**. Despite efforts, the application of AI in drug development still faces several barriers, such as suboptimal model performance due to non-diverse training data and shortcut learning or the introduction of new risks associated with complex, opaque model architectures. Both model development and deployment must address these risks to ensure patient safety and the integrity of clinical study results.

Moreover, the data-driven nature of AI needs active measures to foster trustworthiness. The non-always transparent decision-making processes of AI systems can conflict with regulatory frameworks that require transparent rationales for decisions. To achieve trustworthy AI, several requirements, such as technical robustness and safety, accountability, non-discrimination, and fairness, need to be applied to the AI systems life cycle. Overall, **the evidence reviewed underscores the importance of ensuring that AI applications throughout the medicinal product lifecycle comply with existing legal**

requirements, ethical principles, and fundamental rights, which highlights that special attention should be given to those requirements impacting individuals and the need to evaluate the requirements prospectively.

Below, guiding principles and good practices are provided for the general application of AI in the context of drug development:

- **Multidisciplinary expertise throughout the product lifecycle:** A comprehensive understanding of how a model integrates into clinical workflows, alongside its intended benefits and associated patient risks, is essential. This ensures that ML tools are both safe and effective, addressing clinically significant needs throughout the device's lifecycle.
- **Implementation of security practices:** Adopting robust software engineering practices, ensuring data quality, and implementing strong cybersecurity measures are crucial. These practices involve risk management and design processes that effectively capture and communicate design decisions, implementation strategies, and risk management rationale while maintaining data authenticity and integrity.
- **Clinical study participants:** Data collection protocols must ensure that the characteristics of the intended patient population—such as age, gender, sex, race, and ethnicity—are adequately represented in clinical studies and training/test datasets. This ensures results can be generalised to the target population, managing bias, promoting generalisable performance, and identifying conditions where the model may underperform.
- **Model design and data availability:** Model design should align with the available data and actively address known risks, including security concerns. A thorough understanding of the clinical benefits and risks associated with the drug should inform the development of clinically relevant performance goals, ensuring the product can achieve its intended use safely and effectively.
- **Testing under clinically relevant conditions:** Statistical plans should be developed and executed to evaluate drug performance under conditions that are independent of the training dataset. This involves considerations of the intended patient population, important subgroups, clinical environment, AI techniques, measurement inputs, and potential confounding factors.
- **Clear information for users:** Users should have access to clear and contextually relevant information appropriate for their audience—such as healthcare providers or patients. This includes details on the medicine's intended use, model performance for various subgroups, characteristics of the training and testing data, known limitations, user interface interpretation, and integration into clinical workflows.

This study concludes that adopting AI-based tools to support the development of MCM can significantly enhance the EU's crisis preparedness and response capabilities by optimising drug development, improving the efficiency of clinical trials and pre-clinical research, and leveraging the broad benefits of AI in automating and streamlining these processes. AI has demonstrated its ability to reduce time and cost by automating intensive processes, shortening development timelines, and lowering R&D expenditures. Additionally, AI has improved success rates by enabling the early prediction of potential failures and reducing dropout rates in drug discovery and clinical trials. It has also enhanced safety and efficacy by providing advanced analysis and prediction of adverse effects, resulting in the development of safer drugs and vaccines with a higher likelihood of regulatory approval. Ultimately, AI plays a critical role in accelerating innovation, improving therapeutic outcomes, and significantly boosting resource efficiency across the pharmaceutical industry.

This report, while covering topics beyond COVID-19, highlights that much of the recent research on AI for MCM has been spurred by pandemic-driven investments, particularly during the early phases of the medicinal development lifecycle. Furthermore, consultation activities underscored the need for increased funding within the EU to support the development of in-house algorithms, promote

collaboration with industry, and deploy high-quality computational methods across Europe. Notably, the AI tools identified in this report have not been funded by the EU. To remain competitive in the global AI landscape, Europe must carefully balance regulation and innovation, ensuring that stringent regulations do not stifle progress in member states. Achieving this balance is essential for fostering AI advancements while upholding ethical standards and maintaining public trust.

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Annex 1 Methodology

This section outlines the methodology followed in this report for the review of the literature on the use of AI-based tools to support MCM development. Moreover, this section also describes the methodology used to gather additional inputs on the topic from key relevant stakeholders, covering from the selection of relevant stakeholders up to the analysis of the inputs gathered.

Literature review

The systematic literature review was intended to analyse the literature referring to AI-based tools and systems applied to MCM development, enhancing drug discovery, vaccine development, and both pre-clinical and clinical research by accelerating the identification of drug candidates, optimising molecules formulations, and improving trial design, ultimately making the entire process more efficient and effective. An extensive literature search was conducted following the PRISMA guidelines and using the academic databases PubMed, and Google Scholar. **Box *8*** below presents the search string used for the systematic literature review, which was developed as a result of several discussions among the team involved in the study and tested to ensure valid and reliable outcomes were obtained. Studies published within the last 4 years (2020-2024) were considered, and all eligible studies had to be open source, the full text available, and be written in English. The eligibility period covers the past four years since AI has significantly contributed to the development and discovery of new drugs in response to the COVID-19 pandemic. This outbreak led to an explosive need for medical countermeasures.

Box 8. Query string

Titles and abstracts of articles published between 2020 and 2024 were searched for the following terms in PubMed:
("AI" OR "artificial intelligence*" OR "ML" OR "Machine Learning" OR "Deep Learning" OR "computer model*" OR "computational experimentation") AND ("medicine development" OR "MCM development" OR "drug discovery" OR "drug candidate" OR "drug repurposing" OR "vaccine design" OR "vaccine antigen discovery" OR "therapy development" OR "drug development" OR "clinical trial") AND ("outbreak" OR "health emergency" OR "pandemic" OR "infectious disease") AND ("rapid" OR "early" OR "fast" OR "real-time")

Search results were downloaded, resulting in one file prepared for the screening of the search hits (n=283). References were screened by title and abstract to identify whether the studies mentioned AI-based tools or systems (or other emerging autonomous technologies) that are or could be used to 1) accelerate the process of developing medicines relevant to health emergencies (i.e., preclinical and clinical research); 2) support in the development of clinical trials of new medicines (e.g. patients' recruitment, criteria eligibility, prediction and evaluation of outcomes etc.); 3) support in the analysis of critical data for evaluating the safety and effectiveness of a new drug, in particular in preparedness times and during health emergencies; 4) support in rapidly adjusting medicines to new outbreak pathogens; and 5) describe the availability of the identified AI-tools within the literature. A total of 215 publications were selected for further full-text review.

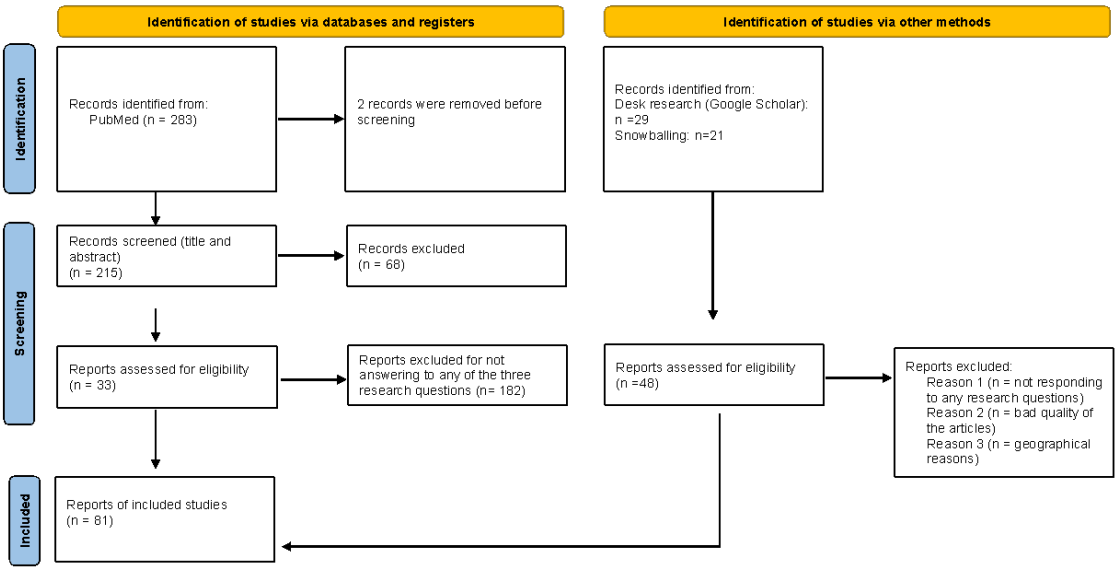
A full-text review was conducted before potentially eligible reports were further assessed. Studies were excluded if they were: 1) not peer-reviewed or from a reputable institution, 2) outside of the research scope, and 3) non-relevant to the research questions. A total of 33 studies were selected for data extraction. Reasons for exclusion for the other studies were recorded. The study team manually added 21 key references identified outside of the database search, which emerged from checking the

references of relevant publications and was considered key (snowballing approach). A single reviewer extracted data from each included report (n=81). A second reviewer checked extractions and appraisals for accuracy and completeness. A narrative synthesis was performed.
The total number of records after each screening round was documented using the PRISMA flow diagram template (**Figure 4**) (PRISMA 2024). The review of the literature allowed the identification of some authors that explored AI-based tools focused on MCM development and in particular the ability to:

- Advance the **development of medicines** and accelerate **drug discovery** relevant for health emergencies.
- Contribute to the efficiency of **clinical trials** in preparedness times and during health emergencies.
- Support the **analysis of critical data**.

Note: This report identifies AI-based tools primarily focused on health emergency preparedness and response. While other AI-based tools could be applicable in this field, the report specifically highlights those designed to support medical countermeasure (MCM) development in the first instance.

Figure 4. PRISMA flow diagram summarising the records of the literature review.



Consultation activities

The consultation activities were carried out for two months and aimed to achieve three primary objectives:

- **Validate and expand upon the research gaps** identified in the literature by incorporating the experiences and perspectives of experts from academia, industry, and government.
- **Identify and prioritise the most crucial challenges** to the adoption of AI in crisis preparedness and emergency response in drugs and vaccine development as well as those related to regulatory affairs.
- **Identify of the most promising AI methods** from a technical perspective and from a clinical perspective to understand future needs and possibilities.

To meet these objectives, the consultations included in-depth interviews of 45-60 minutes long. Before the consultations, three distinct interview guidelines were developed, each tailored to a specific stakeholder category (academia, industry, and government/public health). These guidelines were reviewed and validated by HERA, ensuring comprehensive coverage of key topics while allowing flexibility for individual expertise.

Each guideline included the following sections:

- Introduction and consent to recording and minutes
- General questions about the interviewee's profile and expertise in the field
- Questions related to the AI tools in the 6 main research areas (drug discovery and repurposing, drug target identification, vaccine development, pre and clinical research, and decision-making in health crisis preparedness and response)
- Questions related to challenges and limitations, and
- Questions about additional sources or experts to consult.

A comprehensive stakeholder mapping exercise was conducted at the outset of the study (and lasted four months), identifying key experts across the four stakeholder categories. This mapping was validated by HERA and informed the subsequent engagement activities.

The consultation strategy envisioned a target number of participants for each category:

- Academia (N = 5)
- Industry (N = 5)
- Government (N = 5)

During the two-month consultation period, other stakeholders from those originally selected were included in the interviews, after a thorough review of their expertise. **Table 6** below summarises the number of stakeholders contacted and the rate of positive answers.

Table 6. Contacted stakeholders for the interviews

	Interview (N)	Positive answer (%)
Academia	25	20%
Industry	15	27%
Government	21	24%

Ultimately, **5** stakeholders from academia, **4** from industry, and **5** from government were consulted, providing invaluable feedback and input.

The interviews were conducted in a semi-structured manner, leaving room for further tailoring to the interviewee's expertise. Approximately one week before the interview, some preparatory materials were shared with the interviewees, containing preliminary questions.

The valuable insights gathered from the consultation activities are presented at the end of each relevant results section (Boxes 2 to 7). These findings collectively contributed to the final version of the limitations and discussion sections.

This scientific background document has been peer-reviewed by three experts, and their comments and suggestions were incorporated into the final version.

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